Finite Excess in Ordinal Nim Multiplication: Four Selected-Order Problems

GPT 5.6 Sol Claude Fable 5 a9lim

Abstract

Lenstra's construction of ordinal nim multiplication assigns a finite excess m_p to every odd prime p. The observed values obey a simple $0/1/4$ rule determined by the component support of $f(p) = \mathrm{ord}_p(2)$, but no universal proof is known.

We give an exact status theorem. The rule is equivalent to four full-primary-order assertions for elements selected by the literal Conway tower: a synchronized structural norm, a marked cyclotomic value, a Gaussian period in a norm-one torus, and a corrected conductor-five norm. These are lossless reductions, not hypotheses, and all four universal assertions remain open. The paper then isolates the arithmetic content of each coordinate. It constructs the primitive-support quotient and the Conway–Fermat packet recursion and a packet-degree defect bound, proves that the Conway–Fermat coordinate and Wiedemann's iterated quadratic tower select generator candidates of the same Fermat torus that are never conjugate from level three on, identifies the ordinary Artin symbol, proves least-counterexample one-primary regimes in the Fermat and ordinary arms, the cubic Singer recursion, a shared minimal-coefficient-field packing bound for the ordinary and cubic arms, and a partition selected-order bound, and a sharp first-relation threshold with a single reciprocal section and an exact power-state composition saturation theorem and transverse first-jet boundary, classifies the selected top-orbit elliptic S-unit lattice, proves an ordinary section sieve, and gives the exceptional Capelli criterion, a primitive-only palindromicity test, a one-third Frobenius compression, a Cayley-coordinate norm boundary, an intermediate-field packing sieve, and a level-dependent translate bound. The compression proves the exceptional arm whenever its current cyclotomic quotient is prime, and the packing sieve proves a selected composite large-factor range, while the remaining composite levels stay open. A final reciprocity theorem shows that Frobenius, character, and affine-holonomy transforms of one marked Kummer phase can only erase or re-encode that phase; they cannot evaluate it without an independent ancestry input.

The proof record distinguishes paper theorems, ordinary kernel-checked algebra, native-decided named finite certificates, source-backed data, and the still-open universal rule.

1 Introduction

Conway's nim addition and multiplication make the ordinals a proper class field of characteristic two. Its finite subfields are canonically embedded, and its first transfinite layers are assembled by explicit Kummer extensions [4, 9, 10]. Lenstra's description associates to each odd prime p an element

$$\alpha_p = \kappa_{f(p)} + m_p, \qquad f(p) = \mathrm{ord}_p(2),$$

where $\kappa_{f(p)}$ is determined by the tower and $m_p \in \mathbb{N}$ is the *finite excess*. The same structure appears in the description of the ordinal fields On_p [6].

Let $\mathcal{Q}(h)$ be Lenstra's component set for κ_h. The available data suggest the following rule.

Conjecture 1.1 (the 0/1/4 rule)**.** *For every odd prime p,*

$$m_p = \begin{cases} 0, & \mathcal{Q}(f(p)) \text{ is not a singleton odd prime power,} \\ 4, & f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

The term *selected* is essential. A finite field contains many elements having the same degree, trace, norm, or cyclotomic support, whereas the Conway construction recursively marks one Frobenius orbit. The open content is the primary multiplicative order of that orbit. Computations are valuable for individual rows and as countermodels to overstrong lemmas, but they do not prove Conjecture 1.1.

Theorem A. *The 0/1/4 rule is equivalent to the conjunction of four selected-order assertions Z, O, C, D stated in Theorem 3.1. Each assertion is an exact Euler nonpower test for one Conway-selected coordinate and retains the full primary valuation. The reductions, the coordinate formulas, the packet-universal Fermat residual of Proposition 4.7, the packet-degree bound of Proposition 4.5, the least-counterexample packet sharpening of Corollary 4.6, the Wiedemann-sibling comparison of Proposition 4.14, the all-iterate ramification boundary of Proposition 4.10, the generalized-Prym saturation boundary of Proposition 4.11, the selected top-orbit S-unit reduction of Proposition 4.13, selected Singer, ratio-packing, and partition bounds of Propositions 6.3, 6.4, and 6.6, the cubic first-relation, power-state composition, first-jet, and marked-projective obstructions of Propositions 6.7, 6.8, 6.9, and 6.10, and the exceptional compression, packing, and translate bounds of Propositions 7.6, 7.7, 7.8, and 7.9 are proved here. Corollary 5.4 settles the ordinary large-factor case, Proposition 5.5 doubles its degree range whenever the literal bottom polynomial is not self-reciprocal, and Proposition 5.9 enlarges that range exponentially in $\sqrt{r^a}$ for fixed r, while Corollary 5.11 isolates a one-primary least-counterexample range; the compression settles the exceptional prime-complement case, and the intermediate-field sieve settles an exponential-in-$\sqrt{3^k}$ composite large-factor range. None of Z, O, C, D is proved universally; consequently the 0/1/4 rule and absolute boundedness of the finite excess remain open.*

This theorem is the organizing spine of the paper. The point is not to replace one conjecture by four opaque ones, but to remove every ambient-field ambiguity while preserving the one datum that coarse invariants lose: the Frobenius orbit selected recursively by the Conway tower.

After finite-field preliminaries, Section 3 gives the four-arm equivalence. Sections 4–7 develop one lossless selected coordinate for each arm. Section 8 proves the saturation boundary for reciprocity arguments built from one marked phase. Section 9 separates the stronger 0/1/4 rule from absolute boundedness, and Section 10 records the authority of every claim class used in the paper.

2 Finite-field preliminaries

For a finite algebraic nimber x, write

$$d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2].$$

The element κ_h is the sum of the selected components κ_q, $q \in \mathcal{Q}(h)$, and $h \mid d(\kappa_h)$. The excess has the equivalent form

$$m_p = \min\{m \geq 0 : \kappa_{f(p)} + m \text{ is not a } p\text{-th power in its defining finite field}\}.$$

Finite integers are identified with their canonical finite nimbers. We use standard facts about finite fields and Capelli composition [11].

Proposition 2.1 (exact power criterion)**.** *Let p be prime, let $p \mid 2^E - 1$, and let $x \in \mathbb{F}_{2^E}^{\times}$. The following are equivalent:*

1. *x is a p-th power;*
2. *$x^{(2^E-1)/p} = 1$;*
3. *$\mathrm{ord}(x) \mid (2^E - 1)/p$.*

Consequently, x is not a p-th power if and only if

$$v_p(\mathrm{ord}(x)) = v_p(2^E - 1).$$

Proof. Choose a generator g of the cyclic multiplicative group and write $x = g^a$. Solvability of $py \equiv a \pmod{2^E - 1}$, the displayed power test, and the order divisibility are equivalent. The final statement is their p-primary form. □

The valuation cannot be replaced by the weaker assertion $p \mid \mathrm{ord}(x)$. If $h = f(p)$ and $h \mid E$, lifting the exponent gives

$$v_p(2^E - 1) = v_p(2^h - 1) + v_p(E/h).$$

Every order assertion below retains the full right-hand valuation.

Proposition 2.2 (norm reduction)**.** *Let $h \mid E$, let $p \mid 2^h - 1$, and let $x \in \mathbb{F}_{2^E}^{\times}$. Then*

$$x^{(2^E-1)/p} = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x)^{(2^h-1)/p}.$$

Thus x is a p-th power in $\mathbb{F}_{2^E}$ if and only if its norm is a p-th power in $\mathbb{F}_{2^h}$.

Proof. The norm exponent is $(2^E-1)/(2^h-1)$. Multiply it by $(2^h-1)/p$ and apply Proposition 2.1. □

3 Exact four-arm reduction

For $h > 1$, put

$$\Pi_h = \prod_{\substack{\ell \text{ odd prime} \\ \mathrm{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}, \qquad T_h = \mathrm{N}_{\mathbb{F}_{2^{d(\kappa_h)}}/\mathbb{F}_{2^h}}(\kappa_h).$$

Thus Π_h is the full primary part supported at primes first occurring at residue degree h.

For an odd prime r, let $b_r = d(\alpha_r)$. Put $x_{r,0} = \alpha_r$, and for $a \geq 1$ put

$$x_{r,a} = \kappa_{r^a}.$$

For every $a \geq 0$, define

$$F_{r,a} = \mathbb{F}_{2^{r^a b_r}}, \qquad y_{r,a} = x_{r,a} + 1,$$

and for $a \geq 1$ define

$$G_{r,a} = F_{r,a}^{\times}/F_{r,a-1}^{\times}, \qquad L_{r,a} = |G_{r,a}| = \Phi_{r^a}(2^{b_r}).$$

For $k \geq 1$, set $h = 3^k$, let $\zeta_k = \kappa_{3^k}$, and put

$$\gamma_k = \zeta_k + \zeta_k^{-1} \in \mathbb{F}_{2^h}^{\times}.$$

Finally, let $q = 2^h$, let $\omega = \zeta_k^h \in \mathbb{F}_4$, and define

$$A_k = \zeta_k^2 + \zeta_k + \omega, \qquad M_k = A_k^{q-1}, \qquad \Psi_k = \frac{\Phi_{2\cdot 3^k}(2)}{3}.$$

Theorem 3.1 (four-arm reduction)**.** *Conjecture 1.1 is equivalent to the conjunction of the following four universal assertions.*

Z. If $\mathcal{Q}(h)$ is not a singleton odd prime power, then $\Pi_h \mid \mathrm{ord}(T_h)$.

O. If $r \neq 3$ is odd, $a \geq 1$, and $p \neq r$ is a prime divisor of $L_{r,a}$, then

$$v_p\Big(\mathrm{ord}(y_{r,a}F_{r,a-1}^{\times})\Big) = v_p(L_{r,a}).$$

C. The element γ_k is primitive in $\mathbb{F}_{2^{3^k}}^{\times}$ for every $k \geq 1$.

D. One has $\Psi_k \mid \mathrm{ord}(M_k)$ for every $k \geq 1$.

Proof. Fix p and put $h = f(p)$. If $\mathcal{Q}(h)$ is not a singleton odd prime power, Proposition 4.1 below identifies $m_p = 0$ with the corresponding primary factor of Z.

If $\mathcal{Q}(h) = \{r^a\}$ with r odd, Lenstra's component formulas give $h = r^a s$ with $s \mid b_r$. Proposition 5.1 identifies the row primes with $p \neq r$ dividing $L_{r,a}$ and identifies $m_p = 1$ with assertion O.

When $r = 3$, one has $b_3 = 2$. The cases are $h = 3^a$ and $h = 2 \cdot 3^a$. Proposition 6.1 identifies the former with C, while Theorem 7.3 identifies the latter with D. These cases are disjoint and exhaustive. □

Remark 3.2. Theorem 3.1 is an equivalence, not a proof of any of its four universal assertions. None of Z, O, C, D is presently known in full.

4 The zero arm: structural norms and Fermat packets

4.1 Synchronized structural norm

Proposition 4.1 (exact support criterion)**.** *Let $E = d(\kappa_h)$. For an odd prime p with $f(p) = h$,*

$$m_p = 0 \quad \Longleftrightarrow \quad v_p(\mathrm{ord}(T_h)) = v_p(2^h - 1).$$

Consequently every such row has excess zero if and only if $\Pi_h \mid \mathrm{ord}(T_h)$.

Proof. By the defining excess formula, $m_p = 0$ exactly when κ_h is not a p-th power. Proposition 2.2 moves the test to T_h, and Proposition 2.1 gives the primary valuation. □

When κ_h has several components, T_h is not determined by separate component norms. Their common embedding carries a synchronized Frobenius phase. The following quotient removes every proper-subfield ambiguity while retaining that phase.

Proposition 4.2 (cubical primitive-support coordinate)**.** *Let S be the set of prime divisors of h, let $M = \mathbb{F}_{2^h}$, and write $\varphi(x) = x^2$ and $\sigma_r = \varphi^{h/r}$. For $A \subseteq S$, put $\sigma_A = \prod_{r \in A} \sigma_r$, and define*

$$H_h = \prod_{r \in S} \mathbb{F}_{2^{h/r}}^{\times}, \qquad \Delta_h(x) = \frac{\prod_{|A| \text{ even}} \sigma_A(x)}{\prod_{|A| \text{ odd}} \sigma_A(x)}.$$

Then

$$\ker \Delta_h = H_h, \qquad |\operatorname{im} \Delta_h| = \Phi_h(2),$$

and Δ_h induces an isomorphism $M^{\times}/H_h \simeq \operatorname{im} \Delta_h$. Arm Z at degree h is equivalent to

$$\Pi_h \mid \mathrm{ord}(\Delta_h(T_h)).$$

Proof. The operator is the power map

$$x \longmapsto x^{\prod_{r\mid h,\ r \text{ prime}}(1-2^{h/r})}.$$

In the cyclic group of order 2^h-1, lifting the exponent prime by prime shows that its kernel has order $\operatorname{lcm}_{r\in S}(2^{h/r}-1)$, exactly the order of H_h. The cyclotomic product formula gives quotient order $\Phi_h(2)$. A prime with residue degree h divides no proper-subfield order, so the quotient preserves its full primary valuation. Proposition 4.1 gives the last assertion. □

The cubical coordinate is canonical, but nonidentity is weaker than full primary order. Moreover, an additive cube made from independently shifted components is not the synchronized norm of their selected sum. Thus neither linear disjointness nor a nonzero raw cross-ratio proves Z.

4.2 The singleton-even Conway–Fermat chain

The zero arm also contains the singleton-even chain. Put

$$h_n = 2^{n+1}, \qquad c_n = \kappa_{h_n} = 2^{2^n}, \qquad E_n = \mathbb{F}_{2^{h_n}}, \qquad F_n = 2^{2^n}+1,$$

and $E_{-1} = \mathbb{F}_2$. The tower relation is

$$c_n^2 + c_n = \prod_{j<n} c_j.$$

Let

$$\delta_n = \operatorname{ord}(c_n E_{n-1}^\times) \quad \text{in } E_n^\times/E_{n-1}^\times.$$

Theorem 4.3 (Conway–Fermat selected order)**.** *One has $1 < \delta_n \mid F_n$, and the zero arm at every prime divisor of F_n is equivalent to*

$$\delta_n = F_n \qquad (n \geq 0).$$

All compatible binary root choices form one absolute-Frobenius orbit, so they have the same order vector $(\delta_0, \dots, \delta_n)$. The remaining assertion is therefore intrinsic to the selected tower rather than to an extra choice of root.

Proof. The quotient group has order $(2^{2^{n+1}}-1)/(2^{2^n}-1) = F_n$, and $c_n \notin E_{n-1}$. Every prime $p \mid F_n$ has $f(p) = 2^{n+1}$, with its full primary part in this quotient. Proposition 2.1 gives the displayed equality for δ_n.

A compatible prefix has two extensions at each level, so there are 2^{n+1} compatible tuples. Frobenius preserves the defining equations, and the 2^{n+1} Frobenius conjugates are distinct because c_n has that degree. They exhaust the tuples and preserve all relative orders. □

The quotient in Theorem 4.3 is a torus. Put

$$q = 2^{2^n} = |E_{n-1}|, \qquad T_n = \ker\bigl(N_{E_n/E_{n-1}} \colon E_n^\times \to E_{n-1}^\times\bigr), \qquad u_n = \frac{c_n+1}{c_n},$$

and $a_{n-1} = \prod_{j<n} c_j$, with $a_{-1} = 1$.

Proposition 4.4 (Fermat torus coordinate)**.** *The group T_n is cyclic of order F_n, and $z \mapsto z^{q-1}$ induces an isomorphism $E_n^\times/E_{n-1}^\times \simeq T_n$ sending $c_n E_{n-1}^\times$ to u_n. Moreover*

$$u_n + u_n^{-1} = a_{n-1}^{-1}, \qquad a_{n-1} = \frac{u_n}{(u_n+1)^2}.$$

Consequently $\mathrm{ord}(u_n) = \delta_n$*, and the Conway–Fermat equality $\delta_n = F_n$ says exactly that u_n generates T_n.*

Proof. The group $E_n^\times$ is cyclic of order q^2-1 and $N_{E_n/E_{n-1}}(z) = z^{q+1}$, so T_n is its subgroup of order $q+1 = F_n$. The $(q-1)$-st power map has kernel $E_{n-1}^\times$ and image T_n. The generator $z \mapsto z^q$ of $\mathrm{Gal}(E_n/E_{n-1})$ fixes a_{n-1}, hence permutes the two roots c_n and c_n+1 of X^2+X+a_{n-1}; since $c_n \notin E_{n-1}$, it exchanges them. Thus $c_n^{q-1} = (c_n+1)/c_n = u_n$. Finally

$$u_n + u_n^{-1} = \frac{(c_n+1)^2 + c_n^2}{c_n(c_n+1)} = \frac{1}{c_n^2+c_n} = \frac{1}{a_{n-1}},$$

and $u_n/(u_n+1)^2 = (u_n+u_n^{-1})^{-1}$ in characteristic two. □

The recurrence has a deterministic packet form. For each divisor $e > 1$ of F_n, the inverse pairs of primitive e-th roots give a squarefree packet polynomial

$$C_e(X) = \prod_{[z]\in\mu_e^{\mathrm{prim}}/(z\sim z^{-1})} \left(X - \frac{z}{(z+1)^2}\right) \in \mathbb{F}_2[X].$$

Every irreducible factor has degree 2^n, and

$$\delta_n = e \quad\Longleftrightarrow\quad \text{the selected minimal polynomial of } a_{n-1} \text{ divides } C_e.$$

If P is such a factor, its unique irreducible child is

$$\mathrm{Res}_X(P(X), Y^2 + XY + X^3).$$

Thus the literal chain is one deterministic orbit beginning at $X+1$. Full parent conductor does not imply full child conductor: exact degree-16 packet examples have identical full parent order but deterministic children of different conductor. Consequently any induction must use the complete marked orbit, not only the integer conductor.

The packet degree nevertheless gives an unconditional restriction on every possible loss of conductor.

Proposition 4.5 (packet-degree defect bound)**.** *For every $n \geq 0$ one has*

$$\varphi(\delta_n) \geq 2^{n+1}.$$

For $n \geq 2$ the sharper bound is

$$\varphi(\delta_n) \geq 2^{n+2}.$$

If a prime $p \mid F_n$ is absent from the selected order at its full valuation, then

$$p \leq 1 + 2^{2^n-n-1}.$$

For $n \geq 2$ this improves to

$$p \leq 1 + 2^{2^n-n-2}.$$

Proof. The packet C_{δ_n} has degree $\varphi(\delta_n)/2$, while it contains the selected minimal polynomial of a_{n-1}, of degree 2^n. This proves the first inequality. Put $H = 2^{n+1}$. If $n \geq 2$, choose a prime $q \mid \delta_n$; such a prime exists because the first inequality makes $\delta_n > 1$. Since $q \mid F_n$, the order of 2 modulo q is exactly H. Thus $q - 1 = kH$ for some k. Moreover $q \equiv 1 \pmod 8$, so the supplementary law for 2 and Euler's criterion give

$$1 = 2^{(q-1)/2} = \left(2^{H/2}\right)^k = (-1)^k \pmod q.$$

Consequently k is even, every prime divisor of F_n is congruent to 1 modulo $2H = 2^{n+2}$, and $\varphi(\delta_n) \geq \varphi(q) = q - 1 \geq 2^{n+2}$.

Write $a = v_p(F_n)$ and $b = v_p(\delta_n) < a$. Since $\delta_n \mid F_n$, the ratio

$$\frac{\varphi(F_n)}{\varphi(\delta_n)}$$

is an integer. The factor in its prime-by-prime product indexed by p is $p^{a-1}(p-1)$ if $b = 0$, and p^{a-b} if $b > 0$; in either case the ratio is at least $p - 1$. Hence

$$(p-1)2^{n+1} \leq \varphi(F_n) \leq F_n - 1 = 2^{2^n},$$

which is the asserted bound. Replacing 2^{n+1} by the sharper lower bound 2^{n+2} gives the final claim when $n \geq 2$. □

There is a useful least-counterexample sharpening of this bound. Congruence spacing separates a repeated missing coordinate from every genuinely compound defect.

Corollary 4.6 (minimal-counterexample Fermat congruence regimes)**.** *Suppose that p is the least odd prime at which the 0/1/4 rule fails and that its putative failure lies in the Conway–Fermat part of arm Z at level $n \geq 2$. Put*

$$M = 2^{n+2}, \qquad D = F_n/\delta_n, \qquad A_{<p} = \prod_{\substack{q<p \\ q \mid F_n}} q^{v_q(F_n)}, \qquad G = \max\{M, \varphi(A_{<p})\}.$$

Every prime divisor of D is at least p and is congruent to 1 modulo M. Moreover

$$\varphi(F_n) < Gp(p-1) \quad \Longrightarrow \quad D = p.$$

In the larger range

$$\varphi(F_n) < G(p-1)(p+M-1) \quad \Longrightarrow \quad D = p \quad \text{or} \quad p^2 \mid D.$$

If $v_p(F_n) = 1$, the second alternative is impossible and the conclusion in the larger range is again $D = p$.

Proof. Every prime $q \mid D$ is a missing coordinate at the same residue degree. If $q < p$, the row at q would already violate the rule, contrary to minimality. Hence D is p-rough and $p \mid D$.

The proof of Proposition 4.5 shows that every prime divisor of F_n is 1 modulo M. Least-prime minimality also forces the full q-primary part of F_n into δ_n for every $q < p$. Hence $A_{<p} \mid \delta_n$, so

$$G \leq \varphi(\delta_n).$$

Put $t = \varphi(F_n)/\varphi(\delta_n)$. If $D \neq p$, the prime-by-prime factorization of the totient quotient has either a repeated missing p-factor or a distinct missing prime $q > p$. In the repeated case its local factor is at least $p(p-1)$. In the distinct case congruence spacing gives $q \geq p + M$, and the two local factors give

$$t \geq (p-1)(q-1) \geq (p-1)(p+M-1) \geq p(p-1).$$

Multiplication by $\varphi(\delta_n) \geq G$ proves the first implication. The larger strict bound excludes the distinct-prime case and leaves only $D = p$ or $p^2 \mid D$. The valuation-one assertion is immediate.

The square survivor is a precise classical boundary: $p^2 \mid F_n$ implies $2^{p-1} \equiv 1 \pmod{p^2}$, so p is a base-2 Wieferich prime. Thus the congruence sieve isolates, rather than silently assumes, the unresolved repeated-factor obstruction. □

The literal multiplication operator retains that selector in a compact two-block form. At the edge $c_n^2 = c_n + a_{n-1}$, let A denote multiplication by a_{n-1} on E_{n-1}. Multiplication by $a_{n-1}c_n$ on $E_{n-1} \oplus c_n E_{n-1}$ has matrix

$$\begin{pmatrix} 0 & A^2 \\ A & A \end{pmatrix}.$$

This identity is kernel-checked. The Lean declaration is

`fermat_selected_product_mul_pair`.

If S_r denotes the characteristic-two Fibonacci polynomial, the genuine level-n target is

$$S_{F_n/\ell}(a_{n-1}) \neq 0 \qquad (\ell \mid F_n \text{ prime}).$$

Equivalently, one applies the same polynomial to multiplication by a_{n-1} on E_{n-1}. Write M_n for the displayed two-block operator. It represents multiplication by $a_n = a_{n-1}c_n$, so it is the recursive interface to the *next*-level target

$$S_{F_{n+1}/\ell}(M_n) \neq 0 \qquad (\ell \mid F_{n+1} \text{ prime}).$$

At the lower index one instead has the automatic exclusion

$$S_{F_n/\ell}(a_n) \neq 0, \quad \text{equivalently} \quad S_{F_n/\ell}(M_n) \neq 0 \qquad (\ell \mid F_n),$$

because a zero would put a_n in E_{n-1}: zero-block propagation reaches S_{F_n}, and the telescoping partial-trace identity then forces $a_n^{|E_{n-1}|} = a_n$, whereas its Frobenius displacement is $a_{n-1} \neq 0$. This wrong-index exclusion is kernel-checked as

`fibPolyValue_quadratic_birth_lower_period_ne_zero`.

It is not the selected-order conjecture. The block formula remains an exact interface to the displayed next-level nonvanishing problem; it does not prove that target.

At the genuine index there is nevertheless a sharper exact residual. Let $d = F_{n+1}/\ell = 2m+1 > 1$ and put $x = a_n$. Characteristic-two doubling and Cassini give

$$S_m(x)^4 + x^{d-2} = S_d(x)S_{d-2}(x),$$
$$S_{m+1}(x)^4 + x^d = S_d(x)S_{d+2}(x).$$

Consequently, over a field,

$$S_d(x) = 0 \quad \Longleftrightarrow \quad S_m(x)^4 = x^{d-2} \qquad \text{and} \qquad S_{m+1}(x)^4 = x^d.$$

The converse is literal: squaring $S_d = S_{m+1}^2 + xS_m^2$ makes the two displayed powers cancel. The pair is strictly sharper than either equation separately. Indeed, in $\mathbb{F}_2[X]$ its two left-hand residual polynomials have monic gcd exactly S_d. To see the only needed coprimality, any common divisor of S_{d-2} and S_{d+2} divides $X^{d-2}S_4 = X^{d-2}$ by the continuant determinant identity, while the positive odd-index polynomial S_{d-2} has constant coefficient one. Thus S_{d-2} and S_{d+2} are coprime, and the two factorizations give the stated gcd.

This exact branch selector still does not close the Conway–Fermat target. If $d - 1 = 2^s g$ with g odd, its lower equation is

$$S_g(x)^{2^{s+1}} = x^{d-2}.$$

Writing $q = |E_n|$, choose K by the exact congruence $2^{s+1}K \equiv d - 2 \pmod{q-1}$. Since $x \neq 0$, $x^{q-1} = 1$, and the 2^{s+1}-power Frobenius is injective, the equation becomes $S_g(x) = x^K$. This is precisely the existing AE4 inverse-exponent seam. The upper equation removes the adjacent Fibonacci branch but, after $S_d(x) = 0$ is imposed, follows by squaring the same endpoint relation twice and using the lower equation; no present ancestry identity contradicts the resulting monomial equality. The two ordinary identities and the field equivalence are kernel-checked as

`fibPolyValue_odd_lower_fourth_factor`,
`fibPolyValue_odd_upper_fourth_factor`, and
`fibPolyValue_odd_eq_zero_iff_fourth_residuals`.

In fact the inverse-exponent seam is not merely compatible with deficient packets: it is a universal identity on every such packet.

Proposition 4.7 (packet-universal Fermat residual)**.** *Let $s, g \geq 1$, let $d = Rg + 1$ with $R = 2^s$, and let $u \neq 0, 1$ satisfy $u^d = 1$ in a field of characteristic two. Put*

$$\phi(u) = \frac{u}{(u+1)^2}, \qquad x = \phi(u), \qquad f = S_g(x), \qquad h = S_{g+1}(x), \qquad B = \frac{x^g}{f^2}.$$

Then $f \neq 0$ and

$$B = \phi(u^g), \qquad B^R = x, \qquad \frac{h}{f} = T_s(B),$$

where $T_s(z) = z + z^2 + \cdots + z^{2^{s-1}}$. Consequently both fourth-power residual coordinates produced by a zero of S_d are identities for every nontrivial d-th root—equivalently, on every packet C_e with $1 < e \mid d$—before the Conway parent is selected.

Proof. The characteristic-two Binet identity is

$$S_j(\phi(u)) = \frac{u^j + 1}{(u+1)^j}.$$

It first gives $S_d(x) = 0$. Since $\gcd(g, d) = 1$ and $u \neq 1$, it also gives $f \neq 0$, and direct substitution gives

$$\frac{x^g}{f^2} = \frac{u^g}{(u^g+1)^2} = \phi(u^g).$$

The map ϕ commutes with every binary Frobenius power and satisfies $\phi(v^{-1}) = \phi(v)$. Since $u^{Rg} = u^{d-1} = u^{-1}$, this proves $B^R = x$.

Put $t = (u^g + 1)^{-1}$. Then $B = t + t^2$, so the partial Frobenius trace telescopes:

$$T_s(B) = t + t^R = \frac{1}{u^g + 1} + \frac{u}{u + 1}.$$

The last equality uses $(u^g + 1)^R = u^{Rg} + 1 = u^{-1} + 1$. A second application of the Binet identity gives

$$\frac{h}{f} = \frac{u^{g+1} + 1}{(u + 1)(u^g + 1)} = \frac{1}{u^g + 1} + \frac{u}{u + 1},$$

proving the assertion. Thus these residual coordinates characterize the ambient torsion packet but do not distinguish its recursively selected member. □

The Binet identity in this proof is kernel-checked as `fibPolyValue_torus_binet`. The proposition sharpens the failure boundary: any successful induction must use the deterministic parent label in

$$u_{n+1}(u_n + 1)^3 = u_n(u_{n+1} + 1)^2,$$

not only the fourth-power residual, inverse Frobenius, trace, or norm of the child packet.

This isolates the part not settled by the existing Conway-tower literature. Popovych proves lower bounds and verifies the required relative full orders through level eleven using the known Fermat factorizations [15]. Cagliero, Herman, and Szechtman prove an exact product formula for global order in terms of the relative orders, as well as a normal basis theorem for the associated tower elements [2]. Neither result forces every relative factor δ_n to equal F_n; that all-level equality is precisely the remaining selected assertion here.

The limitation of normality and finitely many absolute traces can be made unconditional, independently of the selected tower.

Proposition 4.8 (finite-trace order separation)**.** *For every finite set $S \subset \mathbb{Z}_{>0}$, there are arbitrarily large powers $d = 2^s$ and normal full-degree elements $\alpha, \beta \in \mathbb{F}_{2^d}$ such that*

$$\mathrm{Tr}(\alpha^j) = \mathrm{Tr}(\beta^j) \quad (j \in S), \qquad \mathrm{ord}(\alpha) = 2^d - 1, \quad \mathrm{ord}(\beta) = \frac{2^d - 1}{3}.$$

Consequently no fixed finite tuple of absolute power traces, even together with degree and normality, distinguishes full multiplicative order from a missing factor three in characteristic two.

Proof. Absolute trace is invariant under squaring, so replace every $j \in S$ by its odd core, remove repetitions, and adjoin the exponent one. The resulting odd monomials $X^{u_1}, \ldots, X^{u_r}$ are strongly linearly independent in the sense of Cohen: a nonconstant Artin–Schreier polynomial $H^2 - H + b$ has even leading degree, whereas every nonzero binary linear combination of the chosen monomials has odd leading degree. A rational H cannot evade this argument, since any finite pole would survive in $H^2 - H$.

Apply Cohen's specified-order, specified-functional-trace theorem [3, Theorem 1.1] twice with $q = 2$, first with defect $l = 1$ and then with defect $l = 3$, prescribing the same trace values and prescribing one for the coordinate X. Its explicit degree bound holds for all sufficiently large $d = 2^s$, and $3 \mid 2^d - 1$ for even d. This gives the displayed exact orders and trace equalities.

For an extension of degree a power of the characteristic, an element is normal exactly when its absolute trace is nonzero. By the normal-basis theorem the Frobenius module is free of rank one over the local algebra $\mathbb{F}_2[T]/((T - 1)^d)$, and trace is $(T - 1)^{d-1}$. Thus both elements are normal. The primitive element has full degree. For $d \geq 4$, so does the element of order $(2^d - 1)/3$, since this order exceeds $2^{d/2} - 1$, the largest possible multiplicative order in a proper subfield. □

This proposition is not a counterexample to the Conway–Fermat assertion: it controls generic field elements, whereas δ_n is the order of the torus coordinate attached to the literal recursively selected ancestry. It proves, however, that a successful all-level argument must use more than any fixed finite absolute-trace tuple or the normal-basis theorem. Across a single Conway quadratic edge the entire power-sum recursion is nevertheless explicit: if $y^2 + xy + x^3 = 0$, then

$$y^m + (y + x)^m = D_m(x, x^3),$$

where $D_m(S, B)$ is the signed Dickson–Lucas recurrence with initial values $2, S$ and step $D_{m+2} = SD_{m+1} - BD_m$. This identity is kernel-checked as `conwayChild_powerSum`; it transports trace data but does not recover the missing conductor coordinate.

There is an exact supersingular realization of the selected ancestry.

Proposition 4.9 (supersingular realization and pairing boundary)**.** *On*

$$\mathcal{E}: \quad y^2 + y = x^3 + x^2$$

put $P_n = (c_n, c_{n+1})$. Then $P_n \in \mathcal{E}$, and for $n \geq 2$,

$$[F_n]P_n = O.$$

More precisely, if π is absolute Frobenius and $\iota = \pi + [1]$, then

$$\iota(x, y) = (x + 1, y + x + 1), \qquad \iota^2 = [-1], \qquad \pi^4 = [-4].$$

For $g = (y + 1)/y$, one has

$$g(P_n) = \frac{c_{n+1} + 1}{c_{n+1}}, \qquad \operatorname{ord}(g(P_n)) = \delta_{n+1}.$$

If $T = (0, 0)$, then T has order five and

$$\operatorname{div}(g) = 2[-T] + [2T] - 2[T] - [-2T].$$

Thus g is the quotient of the two fixed doubling Miller functions at $-T$ and T, but it is not a Weil or reduced-Tate pairing function attached to the F_n-torsion point P_n when $n \geq 2$.

Proof. The Conway recursion gives

$$c_{n+1}^2 + c_{n+1} = c_n(c_n^2 + c_n) = c_n^3 + c_n^2,$$

so P_n lies on $\mathcal{E}$. The coordinate formula for $\iota(P) = \pi(P) + P$ follows from the chord law. Applying it twice gives the inverse $(x, y) \mapsto (x, y + 1)$, so $\pi^2 + 2\pi + 2 = 0$ in $\operatorname{End}(\mathcal{E})$ and hence $\pi^4 = [-4]$.

Put $H = 2^{n+1}$. The 2^H-Frobenius fixes c_n and sends c_{n+1} to $c_{n+1} + 1$, hence sends P_n to $-P_n$. For $n \geq 2$,

$$\pi^H = (\pi^4)^{2^{n-1}} = [2^{2^n}],$$

which gives $[2^{2^n} + 1]P_n = O$.

The formula for $g(P_n)$ is immediate from the coordinates, and its order is δ_{n+1} by the definition of u_{n+1}. The five rational points of $\mathcal{E}/\mathbb{F}_2$ show that T has order five. Direct intersection with the two horizontal lines gives

$$\operatorname{div}(y) = 2[T] + [-2T] - 3[O], \qquad \operatorname{div}(y + 1) = 2[-T] + [2T] - 3[O],$$

and subtraction gives the divisor of g. Equivalently, $g = f_{2,-T}/f_{2,T}$ for functions normalized at O. Its divisor is supported on fixed rational five-torsion, with coefficients $2, 1, -2, -1$. For $n \geq 2$, these valuations are not divisible by $F_n \geq 17$, whereas the divisor of an F_n-Miller function and every F_n-th-power correction have all coefficients divisible by F_n. Thus g cannot be such a function attached to P_n, even modulo an F_n-th power. A Weil or reduced Tate pairing on $\mathcal{E}[F_n]$ takes values in μ_{F_n}, whereas $g(P_n)$ has order dividing F_{n+1}. Pairwise coprimality of the Fermat numbers and $\delta_{n+1} > 1$ exclude equality. □

The elliptic curve is only the first stage of a recursive Artin–Schreier tower. Its ramification remains completely rigid at every depth, but that rigidity occurs away from every finite torsion packet.

Proposition 4.10 (all-iterate ramification boundary)**.** *Let $\mathcal{X}_r$ be the smooth projective normalization of the affine curve over $\mathbb{F}_2$ with coordinates $c_0, \dots, c_r$ and equations*

$$c_{j+1}^2 + c_{j+1} = c_j^3 + c_j^2 \qquad (0 \leq j < r).$$

Then $\mathcal{X}_r$ is geometrically connected, the projection $\mathcal{X}_r \to \mathbb{P}^1_{c_0}$ has degree 2^r, and it is unramified over the finite c_0-line. There is a unique point ∞_r above $c_0 = \infty$, and

$$\operatorname{ord}_{\infty_r}(c_j) = -2^{r-j}3^j \qquad (0 \leq j \leq r).$$

Moreover,

$$g(\mathcal{X}_r) = \frac{3^{r+1} - 2^{r+2} + 1}{2}.$$

Consequently no invariant determined only by the branch locus, ramification indices, different, or inertia filtration of the iterated correspondence can distinguish the literal Conway specialization from a deficient finite torsion packet.

Proof. Write $K_j = \overline{\mathbb{F}}_2(\mathcal{X}_j)$. On $\mathcal{X}_0 = \mathbb{P}^1$, the function c_0 has one pole, of order one. Suppose inductively that c_j has a unique pole ∞_j, of order 3^j. The Artin–Schreier driver

$$f_j = c_j^3 + c_j^2$$

then has the same unique pole, of odd order $m_j = 3^{j+1}$; the cubic term has strictly larger pole order than the quadratic term. It cannot equal $w^2 + w$ in K_j: if w has a pole, the leading pole order of $w^2 + w$ is even, and if w has no pole, neither does $w^2 + w$. Thus $Z^2 + Z = f_j$ is irreducible even after extending the constants, proving geometric connectedness and degree two at the next stage.

The reduced Artin–Schreier ramification criterion gives a unique point ∞_{j+1} above ∞_j, with ramification index two and different exponent

$$m_j + 1 = 3^{j+1} + 1.$$

At every other place f_j is integral, and the derivative of $Z^2 + Z - f_j$ is one, so the extension is unramified there. With the normalized valuation upstairs one has $\operatorname{ord}_{\infty_{j+1}}(f_j) = -2m_j$. The defining equation therefore forces

$$\operatorname{ord}_{\infty_{j+1}}(c_{j+1}) = -m_j = -3^{j+1}.$$

Every later quadratic stage doubles the valuation of an earlier coordinate, which proves $\operatorname{ord}_{\infty_r}(c_j) = -2^{r-j}3^j$.

Put $h_j = 2g(\mathcal{X}_j) - 2$. Riemann–Hurwitz gives

$$h_0 = -2, \qquad h_{j+1} = 2h_j + 3^{j+1} + 1.$$

Hence

$$\begin{aligned} h_r &= -2^{r+1} + \sum_{j=0}^{r-1} 2^{r-1-j}\big(3^{j+1} + 1\big) \\ &= 3^{r+1} - 2^{r+2} - 1, \end{aligned}$$

which is the asserted genus formula.

Finally, every nontrivial torsion coordinate u has $c = (u+1)^{-1}$ finite. The literal place $c_0^2 + c_0 + 1 = 0$ and every finite packet specialization therefore meet the same etale locus. The literal place is totally inert through the Conway tower, but that is decomposition data at one marked place, not a consequence of the ramification just computed. The proposition consequently supplies no generator or nonvanishing assertion. □

The anti-invariant generalized Jacobian can also be computed exactly. It is large, but the Kummer function occupies only one tame boundary direction.

Proposition 4.11 (generalized-Prym saturation boundary)**.** *For $r \geq 1$, let*

$$\pi_r : \mathcal{X}_r \longrightarrow \mathcal{X}_{r-1}, \qquad \tau_r(c_r) = c_r + 1, \qquad u_r = \frac{c_r + 1}{c_r}.$$

For $\epsilon = (\epsilon_0, \dots, \epsilon_{r-1}) \in \{0,1\}^r$ and $\delta \in \{0,1\}$, write $P_{\epsilon,\delta}$ for the corresponding binary point of $\mathcal{X}_r$, and put $z(\epsilon) = \#\{i : \epsilon_i = 0\}$. Then

$$\operatorname{div}(u_r) = \sum_{\epsilon \in \{0,1\}^r} 2^{z(\epsilon)}\big([P_{\epsilon,1}] - [P_{\epsilon,0}]\big).$$

In particular its reduced support S_r consists of 2^{r+1} $\mathbb{F}_2$-points, and $B_r = \pi_r(S_r)$ consists of 2^r $\mathbb{F}_2$-points.

Let $J(\mathcal{X}, S)$ denote the reduced-modulus generalized Jacobian. The identity component of the generalized norm kernel

$$\ker\big(J(\mathcal{X}_r, S_r) \longrightarrow J(\mathcal{X}_{r-1}, B_r)\big)$$

has dimension 3^r. Its ordinary Prym quotient has dimension $3^r - 2^r$, while its boundary-torus kernel has dimension 2^r.

For every odd prime ℓ, with $U_r = \mathcal{X}_r \setminus S_r$, the Kummer class

$$\delta_\ell(u_r) \in H^1_{\text{ét}}(U_r, \mu_\ell)$$

comes from the unit term in the Kummer sequence, has tame residue vector given by the displayed divisor formula modulo ℓ, and satisfies $\tau_r^* \delta_\ell(u_r) = -\delta_\ell(u_r)$. This vector is nonzero for every odd ℓ, but its Frobenius orbit spans only one line, up to the cyclotomic Tate twist. The ray character defined by u_r is nontrivial on the anti-invariant boundary torus and is geometrically surjective onto $\mathbb{G}_m$.

Proof. The affine equations are triangular with derivative one in each new coordinate. Hence the binary tuples are distinct smooth points. Conversely, $c_r \in \{0,1\}$ forces

$$c_{r-1}^2(c_{r-1}+1) = c_r(c_r+1) = 0,$$

and induction forces every earlier coordinate to be binary. At a binary point the same identities give

$$\operatorname{ord}(c_i - \epsilon_i) = \begin{cases} 2\operatorname{ord}(c_{i-1}), & \epsilon_{i-1} = 0, \\ \operatorname{ord}(c_{i-1}+1), & \epsilon_{i-1} = 1. \end{cases}$$

Starting with order one on $\mathcal{X}_0$ proves the divisor formula. There is no contribution at infinity, since $u_r = 1 + c_r^{-1}$ there. As a check, each of the zero and pole divisors has degree

$$\sum_{\epsilon \in \{0,1\}^r} 2^{z(\epsilon)} = (2+1)^r = 3^r,$$

the pole degree of c_r.

For a reduced modulus of s geometric points, the boundary torus has dimension $s-1$. The norm on the two boundary points above each point of B_r is pairwise multiplication, so its toric kernel has dimension 2^r. The ordinary Prym dimension is, by Proposition 4.10,

$$g(\mathcal{X}_r) - g(\mathcal{X}_{r-1}) = 3^r - 2^r.$$

The generalized norm is surjective: its composite with pullback is multiplication by two, which is a finite surjective morphism of the semiabelian target even in characteristic two. Subtracting dimensions now gives 3^r for the identity kernel. On prime-to-two Tate modules, the toric norm kernel is exactly the (-1)-part of the boundary-torus module for τ_r; this formulation avoids any false semisimplicity assertion at the characteristic prime.

The Kummer exact sequence gives

$$0 \longrightarrow \mathcal{O}(U_r)^\times / \mathcal{O}(U_r)^{\times \ell} \longrightarrow H^1_{\text{ét}}(U_r, \mu_\ell) \longrightarrow \operatorname{Pic}(U_r)[\ell] \longrightarrow 0.$$

The class of u_r lies in the first term, and its tame residues are its valuations modulo ℓ. All displayed multiplicities are powers of two, so none vanishes modulo an odd ℓ; in particular the reduced modulus loses no conductor information. The coefficient belonging to the all-one prefix is one, proving nonvanishing. Since $\tau_r(u_r) = u_r^{-1}$, the class is anti-invariant and its corestriction is zero.

Every point of S_r is rational over $\mathbb{F}_2$, so Frobenius has no permutation action on the residue vector. Only the scalar cyclotomic twist remains, and the orbit therefore spans one line. Finally, for a degree-zero divisor D disjoint from S_r, put $\chi_{u_r}([D]) = u_r(D)$. If D is changed by $\operatorname{div}(f)$ with $f = 1$ on S_r, Weil reciprocity gives

$$u_r(\operatorname{div}(f)) = f(\operatorname{div}(u_r)) = 1,$$

so this is a character of the generalized Jacobian. Its restriction to the toric norm kernel has the nonzero anti-invariant exponent vector displayed above; a nonzero character of a torus is geometrically surjective onto $\mathbb{G}_m$. □

This surjectivity concerns geometric generalized-divisor classes, not the literal totally inert Conway place. It therefore supplies no full-order claim. Rather, it shows sharply why the ordinary Prym and lower norm data cannot evaluate the selected phase: the relevant ray class is one Frobenius-stable line inside a 2^r-dimensional anti-invariant boundary torus, and selecting its value requires decomposition data at the marked place.

Corollary 4.12 (fixed-five-torsion rank-one boundary)**.** *Let $n \geq 2$, put $A = 2^{2^n}$, $q = A^2$, and let ℓ be an odd prime dividing $q + 1 = F_{n+1}$. Set $L = \mathbb{F}_{q^2}$ and $S = \mathcal{E}(\mathbb{F}_2)$. If $[z]_\ell$ denotes the class of $z \in L^\times$ in $L^\times / L^{\times \ell}$, then*

$$\operatorname{span}_{\mathbb{F}_\ell}\big\{[F(P_n)]_\ell : F \in \mathbb{F}_2(\mathcal{E})^\times, \operatorname{supp}\operatorname{div}(F) \subseteq S\big\} = \mathbb{F}_\ell[g(P_n)]_\ell.$$

Thus no $\mathbb{F}_2$-rational unit supported on the fixed five-torsion supplies an independent Kummer detector for the selected endpoint.

Proof. Write

$$A_0 = [-T] - [T], \qquad B_0 = [-2T] - [2T].$$

Every anti-invariant degree-zero divisor supported on S is $aA_0 + bB_0$. Its class in $\operatorname{Pic}^0(\mathcal{E}) \simeq \mathcal{E}$ is $(3a + b)T$, so the principal lattice is $3a + b \equiv 0 \pmod 5$. Proposition 4.9 gives $\operatorname{div}(g) = 2A_0 - B_0$. With

$$r = f_{5,T} = \frac{y^2(y + x)}{(x + 1)^2}, \qquad s = \frac{r \circ [-1]}{r},$$

one has $\operatorname{div}(s) = 5A_0$; these two divisor vectors have determinant five and hence form a basis of the principal lattice.

For an S-unit F, put $R = (F \circ [-1])/F$. Since $\mathbb{F}_2^\times = \{1\}$, the lattice calculation gives $R = g^m s^k$ for some integers m, k. Direct substitution in the displayed functions gives

$$s = \frac{g^2}{g \circ \iota}.$$

Now $c_n^A = c_n + 1$, so P_n^A and ιP_n have the same x-coordinate. Hence $P_n^A = \epsilon \iota P_n$ for some $\epsilon \in \{1, -1\}$, and

$$s(P_n) = g(P_n)^{\,2 - \epsilon A}.$$

As $P_n^q = -P_n$, it follows that $F(P_n)^{q-1} = R(P_n)$, placing every displayed class on the line generated by $[g(P_n)]_\ell$ because $\gcd(\ell, q - 1) = 1$.

Conversely, for $h = y/(x + 1)$ one has $h \circ [-1]/h = g$. Therefore $h(P_n)^{q-1} = g(P_n)$, and $[h(P_n)]_\ell$ spans the same line. This proves the reverse inclusion. □

The fixed support can be enlarged to the complete top Frobenius orbit without losing the literal selector. The resulting divisor lattice has two explicit function generators after evaluation, but the value quotient still has only one Kummer direction.

Proposition 4.13 (selected top-orbit S-unit reduction)**.** *Let $n \geq 2$ and put*

$$H = 2^{n+1}, \qquad A = 2^{H/2} = 2^{2^n}, \qquad q = 2^H = A^2, \qquad d = 2H, \qquad L = \mathbb{F}_{q^2}.$$

Let $P = (x_P, y_P) \in \mathcal{E}(L)$ satisfy

$$P^q = -P, \qquad [\mathbb{F}_2(x_P) : \mathbb{F}_2] = H, \qquad x_P^A = x_P + 1.$$

These are the top-orbit and half-Frobenius properties of P_n. Let $M \in \mathbb{F}_2[X]$ be the minimal polynomial of x_P, and define the pointwise translated divisors

$$D_0 = \sum_{0 \leq i < d} [P^{2^i}], \qquad D_\pm = \sum_{0 \leq i < d} [P^{2^i} \pm T], \qquad C_0 = D_+ - D_-.$$

Put

$$S' = \mathcal{E}(\mathbb{F}_2) \cup \operatorname{supp}(D_+) \cup \operatorname{supp}(D_-), \qquad f(Q) = M(x(Q+T)), \qquad U = \frac{f \circ [-1]}{f},$$

and

$$\Theta_P = U(P) = \frac{M(x(P-T))}{M(x(P+T))} = M(x(P+T))^{q-1}.$$

Then

$$\operatorname{div}(U) = C_0 + dA_0, \qquad A_0 = [-T] - [T], \quad B_0 = [-2T] - [2T].$$

The $\mathbb{F}_2$-rational, inversion-anti-invariant, degree-zero divisors supported on S' are exactly

$$aA_0 + bB_0 + kC_0 \qquad (a, b, k \in \mathbb{Z}).$$

Such a divisor is principal if and only if

$$3a + b + 2dk \equiv 0 \pmod 5.$$

In the coordinates (A_0, B_0, C_0), the principal lattice has the basis

$$\operatorname{div}(g) = (2, -1, 0), \qquad \operatorname{div}(s) = (5, 0, 0), \qquad \operatorname{div}(U) = (d, 0, 1),$$

where s is the fixed-support function in the proof of Corollary 4.12.

More precisely, if $F \in \mathbb{F}_2(\mathcal{E})^\times$ has divisor supported on S', there are $m, k, r \in \mathbb{Z}$ and $\epsilon \in \{1, -1\}$, with $P^A = \epsilon\iota P$, such that

$$F(P)^{q-1} = g(P)^{\,m+k(2-\epsilon A)}\Theta_P^r.$$

Consequently, for every odd prime $\ell \mid q+1$,

$$\operatorname{span}_{\mathbb{F}_\ell}\big\{[F(P)]_\ell : F \in \mathbb{F}_2(\mathcal{E})^\times, \operatorname{supp}\operatorname{div}(F) \subseteq S'\big\} = \operatorname{span}_{\mathbb{F}_\ell}\{[g(P)]_\ell, [\Theta_P]_\ell\}.$$

This names two function/divisor generators, not two independent value directions: the cyclic quotient $L^\times/L^{\times\ell}$ has $\mathbb{F}_\ell$-dimension one.

Proof. The point P has Frobenius orbit of length d: its x-coordinate has orbit length H, while $P^q = -P \neq P$. The divisors D_+ and D_- are single $\mathbb{F}_2$-orbits. They are disjoint from each other, from $\mathcal{E}(\mathbb{F}_2)$, and from D_0; in particular, $P \notin S'$ and all evaluations below are defined. The orbit D_0 is disjoint from $\mathcal{E}(\mathbb{F}_2)$ by its length, and the same follows for $D_\pm$ after translating by $\mp T$. Any remaining collision would give a nonzero translation of one orbit point by rational five-torsion. Iterating the corresponding Frobenius shift five times would fix that point; since $\gcd(d, 5) = 1$, the original shift would already be trivial, a contradiction.

The roots of M are the x-coordinates of $P, P^2, \ldots, P^{2^{H-1}}$. The two points over each root are interchanged by inversion and together give D_0. Since x has a double pole at O,

$$\operatorname{div}(f) = D_- - d[-T], \qquad \operatorname{div}(f \circ [-1]) = D_+ - d[T].$$

Their difference proves the formula for $\operatorname{div}(U)$. Moreover, $P^q = -P$ and rationality of M, T give

$$M(x(P+T))^q = M(x(-P+T)) = M(x(P-T)),$$

which proves the formula for Θ_P.

Inversion swaps T with $-T$, $2T$ with $-2T$, and the two rational orbits D_+ and D_-. It fixes O, whose coefficient in an anti-invariant integral divisor must therefore vanish. Frobenius invariance makes the coefficient constant on each of D_+ and D_-, proving the asserted rank-three description. Under $\mathrm{Pic}^0(\mathcal{E}) \simeq \mathcal{E}$ one has

$$A_0 \longmapsto 3T, \qquad B_0 \longmapsto T, \qquad C_0 \longmapsto 2dT.$$

This proves the congruence criterion. The three displayed divisor vectors belong to its kernel and have determinant five. Since the congruence kernel itself has index five in $\mathbb{Z}^3$, they form a basis.

For an S'-unit F, set $R = (F \circ [-1])/F$. Its divisor belongs to the principal anti-invariant lattice, so

$$R = g^m s^k U^r$$

for some integers m, k, r; no scalar remains because $\mathbb{F}_2^\times = \{1\}$. The selected half-Frobenius equation gives $x(P^A) = x(\iota P)$, hence $P^A = \epsilon \iota P$ for some $\epsilon \in \{1, -1\}$. Using

$$s = \frac{g^2}{g \circ \iota}, \qquad g(-Q) = g(Q)^{-1},$$

one obtains $s(P) = g(P)^{2-\epsilon A}$. Finally,

$$F(P)^{q-1} = \frac{F(P^q)}{F(P)} = \frac{F(-P)}{F(P)} = R(P),$$

which proves the explicit evaluation. Since $\gcd(\ell, q-1) = 1$, exponentiation by $q-1$ is invertible on $L^\times/L^{\times \ell}$. This gives one inclusion in the span equality; the reverse follows because g and U are themselves S'-units and $U(P) = \Theta_P$. □

The half-Frobenius selector in this proposition is essential. The exact finite certificate in the `experiments` directory is

`fermat_top_orbit_countermodel.py`.

It constructs a generic $H = 32$ point with $P^{2^{32}} = -P$ and x_P of exact degree 32 in

$$\mathbb{F}_2[X]/(\texttt{1c019c923}), \qquad x_P = \texttt{2bf5b2fb}, \qquad M = \texttt{1b3e069a5}.$$

For $\ell = 641$ and $2^{32} + 1 = 641 \cdot 6700417$, it verifies

$$\mathrm{ord}(g(P)) = 6700417, \qquad \mathrm{ord}(\Theta_P) = 2^{32} + 1.$$

Thus $[g(P)]_{641} = 0$ while $[\Theta_P]_{641} \neq 0$. The same certificate finds that the fixed-support value $s(P)$ has full order $2^{32} + 1$, so even the elimination of s fails without the selector. Crucially,

$$x_P^{2^{16}} = \texttt{5e024149} \neq \texttt{2bf5b2fa} = x_P + 1.$$

This is an exact countermodel to any deduction from unmarked top-orbit geometry, not a Conway specialization and not evidence against arm Z. The nonzero classes of $s(P)$ and Θ_P are the same sole 641-direction in the cyclic value quotient; they are not independent Kummer coordinates.

Proposition 4.9 supplies exact CM annihilator control but not generator status. For $n \geq 3$, no elliptic endomorphism can carry P_n to P_{n-1}: the image order would divide both coprime numbers F_n and F_{n-1}, although $P_{n-1} \neq O$. For $n = 2$, the same conclusion follows from $\mathrm{ord}(P_1) = 3$ and

$\mathrm{ord}(P_2) \mid 17$. Thus the relation $x(P_n) = y(P_{n-1})$ is not an isogeny division tower, and pairing functoriality along this proposed tower cannot propagate primitivity. A perfect alternating pairing likewise guarantees some detecting partner, not nonvanishing on a prescribed line.

The tempting cubic curve $Y^2 + Y = X^3$ is not a coordinate rewrite of $\mathcal{E}$: the two curves have respectively three and five rational points over $\mathbb{F}_2$. For $u = (Y+1)/Y$ on that auxiliary curve, $u/(u+1)^2 = X^3$; every positive-level Fermat-torsion value is already a cube because $\gcd(F_n, 3) = 1$. Its natural cubic Kummer class is therefore trivial and cannot replace the selected five-torsion function above. The remaining statement is still exclusion of every proper-conductor packet along the complete resultant chain. Trace, relative norm, irreducibility, and the etale Jacobian hold for proper packets as well as the full packet.

4.3 Wiedemann's sibling tower

Proposition 4.4 places the Conway–Fermat assertion beside an older open problem. Wiedemann's iterated quadratic tower is

$$x_{-1} = 1, \qquad x_m + x_m^{-1} = x_{m-1} \qquad (m \geq 0).$$

Wiedemann proves that every step is quadratic, so that $E_m = E_{m-1}(x_m)$ and x_m^{-1} is the conjugate of x_m over E_{m-1}, and that $x_m^{F_m} = 1$ [22, Theorem 1 and Corollary]. He verifies $\mathrm{ord}(x_m) = F_m$ for $m \leq 8$, asks whether it holds for every m, and, setting his tower beside Conway's, records that no explicit isomorphism between the two presentations was known [22, Sections 4–5]. Seyfarth and Ranade show that the conjecture is equivalent to the assertion that, for every m, the minimal polynomial of x_{m-1}—the m-fold self-reciprocal transform $f \mapsto X^{\deg f} f(X + X^{-1})$ of $X + 1$—has Fibonacci index F_m [17, Theorem 2]. The question remains open. Since $x_{n-1}^{-1} = x_n/(x_n+1)^2$, the Binet identity in the proof of Proposition 4.7 turns it into the residue test

$$S_{F_n/\ell}(x_{n-1}^{-1}) \neq 0 \qquad (\ell \mid F_n \text{ prime}),$$

the Conway target $S_{F_n/\ell}(a_{n-1}) \neq 0$ evaluated at a different selected point.

Both towers have one shape. Call $(b_m)_{m \geq -1}$, with $b_{-1} = 1$, a *trace-compatible quadratic tower* if $b_m \in E_m \setminus E_{m-1}$ and

$$b_m^2 + b_{m-1} b_m + \nu_m = 0, \qquad \nu_m \in E_{m-1}^{\times} \qquad (m \geq 0).$$

Then $\mathrm{Tr}_{E_m/E_{m-1}}(b_m) = b_{m-1}$ and $N_{E_m/E_{m-1}}(b_m) = \nu_m$, and with $q = |E_{m-1}|$ the norm-one coordinate $t_m = b_m^{q-1}$ satisfies

$$t_m + t_m^{-1} = \frac{(b_m^q + b_m)^2}{b_m^{q+1}} = \frac{b_{m-1}^2}{\nu_m}, \qquad \text{that is,} \qquad \frac{t_m}{(t_m+1)^2} = \frac{\nu_m}{b_{m-1}^2}.$$

Proposition 4.14 (Wiedemann sibling)**.** *(a) The sequences (a_m) and (x_m) are trace-compatible quadratic towers, with $\nu_m = a_{m-1}^3$ and $\nu_m = 1$, and with norm-one coordinates u_m and x_m^{-2}. Thus $x_n \in T_n$, and Wiedemann's conjecture at level n says that x_n generates T_n.*

(b) For each m, the compatible tuples $(x_0, \ldots, x_m)$ form one absolute-Frobenius orbit.

(c) The elements u_n and x_n are Frobenius-conjugate if and only if $n \leq 2$.

Proof. (a) The nontrivial automorphism of E_m/E_{m-1} sends c_m to c_m+1, hence $a_m = a_{m-1}c_m$ to $a_{m-1}(c_m+1)$, and it sends x_m to x_m^{-1}. Taking sums and products, and using $c_m(c_m+1) = a_{m-1}$, gives $\mathrm{Tr}(a_m) = a_{m-1}$, $N(a_m) = a_{m-1}^3$, $\mathrm{Tr}(x_m) = x_{m-1}$, and $N(x_m) = 1$. Moreover $a_m \notin E_{m-1}$, because $a_{m-1} \in E_{m-1}^\times$ and $c_m \notin E_{m-1}$. The norm-one coordinates are $a_m^{q-1} = (c_m+1)/c_m = u_m$ and $x_m^{q-1} = x_m^{-2}$.

(b) A compatible tuple is determined by its top entry, and there are 2^{m+1} of them. Every proper subfield of E_m lies in E_{m-1}, which does not contain x_m, so x_m has 2^{m+1} distinct Frobenius conjugates. Frobenius preserves the defining relations, so these conjugates are the top entries of 2^{m+1} distinct compatible tuples, which are all of them.

(c) The map $z \mapsto z + z^{-1}$ commutes with every field automorphism, and its fibres are the pairs $\{z, z^{-1}\}$; on T_n one has $z^{-1} = z^q$, a Frobenius conjugate of z. Since $u_n + u_n^{-1} = a_{n-1}^{-1}$ and $x_n + x_n^{-1} = x_{n-1}$, and every automorphism of E_{n-1} extends to E_n, the elements u_n and x_n are conjugate exactly when a_{n-1}^{-1} and x_{n-1} are. As x_{n-1}^{-1} is a conjugate of x_{n-1}, this happens exactly when a_{n-1} and x_{n-1} are conjugate.

Let $n \geq 3$, and suppose that an automorphism σ of E_{n-1} sends a_{n-1} to x_{n-1}. It preserves E_{n-2} and commutes with $N_{E_{n-1}/E_{n-2}}$, so (a) gives

$$1 = N(x_{n-1}) = \sigma\big(N(a_{n-1})\big) = \sigma(a_{n-2})^3.$$

Hence $a_{n-2}^3 = 1$ and $a_{n-2} \in \mathbb{F}_4 \subseteq E_{n-3}$, contrary to $a_{n-2} \notin E_{n-3}$.

For $n \leq 2$: one has $a_{-1} = x_{-1} = 1$; next, $a_0 = c_0$ and x_0 are both roots of $X^2 + X + 1$; finally, a_1 has trace a_0 and norm $a_0^3 = 1$, so it is a root of $X^2 + a_0X + 1$, which is the Wiedemann relation for the compatible choice $x_0 = a_0$. By (b), a_1 is conjugate to x_1. □

For instance, $a_2^{17} = N(a_2) = a_1^3$ has order five, because a_1 is a root of $X^2 + a_0X + 1$ and hence of $X^4 + X^3 + X^2 + X + 1$, while the image of a_2 in $E_2^\times/E_1^\times$ is that of c_2, of order $\delta_2 = 17$. Thus a_2 has order 85, whereas x_2 lies in T_2, of order 17.

In the trace-compatible coordinates the two recursions therefore share their trace prescription and differ in their norm prescription. On the selected torus elements the parent laws are

$$\frac{x_n}{(x_n+1)^2} = \frac{1}{x_{n-1}}, \qquad \frac{u_n}{(u_n+1)^2} = \frac{u_{n-1}}{(u_{n-1}+1)^3} \qquad (n \geq 1),$$

the second by Proposition 4.4 and $c_{n-1} = (u_{n-1}+1)^{-1}$. The first is the genus-zero correspondence of iterated preimages of $z \mapsto z + z^{-1}$: every entry of a Wiedemann tuple is a rational function of the top entry, so the analogue of the curve $\mathcal{X}_r$ of Proposition 4.10 is the projective line at every depth. The second is birational to the supersingular curve of Proposition 4.9, and its iterates have the growing genus of Proposition 4.10. Proposition 4.7 is independent of the selected ancestry and holds equally for Wiedemann's torus elements, whereas Proposition 4.8 concerns generic field elements and gives no separation theorem inside the Fermat tori or along either selected sequence. The two selections differ only through their parent laws, and part (c) shows that they are distinct from level three on: no isomorphism between Conway's and Wiedemann's presentations of E_n carries u_n to x_n when $n \geq 3$.

Remark 4.15 (ordinary realization). Wiedemann's tower is an isogeny division tower, on a curve of the opposite type. On the ordinary curve

$$\mathcal{E}' : \quad y^2 + xy = x^3 + 1$$

over $\mathbb{F}_2$, let π be absolute Frobenius and V its dual, so that $V\pi = [2]$ and $V = [2]\circ\pi^{-1}$ on geometric points. Duplication gives $x([2]Q) = x(Q)^2 + x(Q)^{-2}$, so

$$x(VQ) = x(Q) + x(Q)^{-1} \qquad (x(Q) \neq 0).$$

Voloch uses this identity to place Wiedemann's elements on $\mathcal{E}'$ [21, Section 4], and Ugolini studies the dynamics of $z \mapsto z + z^{-1}$ through the same abscissa action of the dual Frobenius [20]. Since π preserves orders, $\operatorname{ord}(VQ) = \operatorname{ord}(Q)/\gcd(2, \operatorname{ord}(Q))$. The rational points $(1,0)$ and $(1,1)$ have order four, because $[2](1,0) = (0,1)$ and $[2](0,1) = O$, and they are the only points with abscissa one. Every point of order at least eight has nonzero abscissa, so V^{n+1} carries a geometric point with abscissa a level-n Wiedemann element to a point of order four, and conversely. Hence the level-n Wiedemann elements are exactly the abscissas of the geometric points of exact order 2^{n+3} on $\mathcal{E}'$. Such a point is rational over E_n: with $z = y/x_n$, its ordinate solves $z^2 + z = x_n + x_n^{-2}$, whose absolute trace is $\operatorname{Tr}_{E_n/\mathbb{F}_2}(x_n + x_n^{-1}) = \operatorname{Tr}_{E_n/\mathbb{F}_2}(x_{n-1}) = 0$.

Thus Wiedemann's tower is the abscissa tower of the 2-primary torsion of an ordinary curve, whereas the selected points of Proposition 4.9 are annihilated by the odd number F_n on a supersingular curve and, as shown above, do not form a division tower. Ordinary and supersingular curves are not isogenous, even over $\overline{\mathbb{F}}_2$. This separates the two geometric realizations; it does not exclude other relations between the two multiplicative-order questions.

The ordinary realization carries a published asymptotic lower bound for Wiedemann's tower. Voloch proves $\operatorname{ord}(x_n) \geq \exp(2^{\delta n})$ for an unspecified $\delta > 0$ and all large n [21], and Popovych proves the effective bound $\operatorname{ord}(x_n) \geq 7 \cdot 2^{n+2} + 1$ for $n \geq 12$ together with $\operatorname{ord}(x_n) = F_n$ for $n \leq 11$ [14, Theorem 3]. For the Conway–Fermat chain the published lower bound is the Euler–Lucas congruence bound $\delta_n \geq 2^{n+2} + 1$ for $n \geq 2$ [15, 2], which Proposition 4.5 recovers. Voloch's theorem concerns a fixed function y on an elliptic curve evaluated at a point P whose cyclic group is Frobenius-stable and whose order r is below $[\mathbb{F}_q(P) : \mathbb{F}_q]^{3/2-\epsilon}$ [21, Theorem 4.1]. Its proof chooses many small integers k in one coset $\gamma\langle a\rangle$ of the image of Frobenius in $(\mathbb{Z}/r\mathbb{Z})^\times$; with $C = [\gamma^{-1}]P$, the values $y([k]C)$ are Frobenius conjugates of $y(P)$, and products of them are separated by the multiplicative independence of the functions $y \circ [k]$. On $\mathcal{E}'$ the image of Frobenius is everything up to sign: if Q has exact order 2^{n+3} and abscissa x_n, the 2^{n+1} conjugates of x_n are abscissas of points of exact order 2^{n+3}, and these are the 2^{n+1} abscissas $x([k]Q)$ with k odd. In this specialization the argument is explicit.

Proposition 4.16 (explicit Wiedemann bound)**.** *For every $n \geq 0$,*

$$\operatorname{ord}(x_n) \geq 2^t, \qquad t = \left\lfloor (3 \cdot 2^{n-2})^{1/3} \right\rfloor.$$

Proof. Let Q be as above and put $f_k = x \circ [k]$ for odd $k \geq 1$, a function of degree $2k^2$ on $\mathcal{E}'$. Each $f_k(Q)$ is a conjugate of x_n, hence lies in $\langle x_n\rangle$. The functions f_k are multiplicatively independent: $\operatorname{div}(x) = 2[T] - 2[O]$ with $T = (0,1)$, and for the largest k in a putative relation, a geometric point R of exact order $2k$ is a zero of f_k, whereas for odd $\ell < k$ the point $[\ell]R$ is neither T nor O. If two subset products of $f_1(Q), f_3(Q), \ldots, f_{2t-1}(Q)$ coincide, the quotient of the corresponding functions minus one is a nonzero function over $\mathbb{F}_2$ vanishing at Q, hence at its 2^{n+1} conjugates, while its degree is at most

$$2\sum_{j=1}^{t}(2j-1)^2 = \frac{8t^3 - 2t}{3} < 2^{n+1}.$$

Thus the 2^t subset products are distinct elements of $\langle x_n\rangle$. □

This beats Popovych's effective bound from $n = 13$ on, and it is far from F_n. The selected Conway element also has a fixed-function realization, $u_{n+1} = g(P_n)$ in Proposition 4.9, but its Frobenius image is thin.

Proposition 4.17 (Frobenius-thin Conway torsion)**.** *Let $n \geq 2$, put $h = 2^{2^{n-1}}$, and let r be the order of the point P_n of Proposition 4.9. Then $r \mid F_n$, the cyclic group $\langle P_n \rangle$ is stable under ι and absolute Frobenius, ι acts on it as multiplication by h or by $-h$, and Frobenius acts through the subgroup*

$$H_n = \langle 2 \rangle \cup (h+1)\langle 2 \rangle \subseteq (\mathbb{Z}/r\mathbb{Z})^\times, \qquad |H_n| = 2^{n+2} = [\mathbb{F}_2(P_n) : \mathbb{F}_2].$$

In particular $r > 2^{n+2}$. Suppose $r = F_n$. Then no odd integer k with $1 < |k| < h - 1$ lies in H_n, and for distinct odd positive integers k, k' with $kk' < h/2$ there is no congruence $k' \equiv \alpha k \pmod{F_n}$ with $\alpha \in H_n$; when both are units, they lie in distinct cosets of H_n.

Proof. Proposition 4.9 gives $[F_n]P_n = O$. Let $\sigma = \pi^{2^n}$ be the $|E_{n-1}|$-power Frobenius. It sends c_n to $c_n + 1$, hence $a_n = a_{n-1}c_n$ to $a_n + a_{n-1}$, so it sends c_{n+1} to a root of $Z^2 + Z = a_n + a_{n-1}$, namely $c_{n+1} + c_n$ or $c_{n+1} + c_n + 1$. By the coordinate formula for ι and $-(x, y) = (x, y + 1)$, this says $\sigma(P_n) = \pm\iota P_n$. On the other hand $\pi^4 = [-4]$ gives $\sigma = [(-4)^{2^{n-2}}] = [\pm h]$. Hence $\iota P_n = \pm h P_n$, and since $\pi = \iota - [1]$, Frobenius acts on $\langle P_n \rangle$ as multiplication by $m = \pm h - 1$.

Modulo r one has $h^2 = 2^{2^n} \equiv -1$, so -1 and h lie in $\langle 2 \rangle$, which has order 2^{n+1}. Since $(h+1)^2 \equiv 2h \in \langle 2 \rangle$, the set H_n is a subgroup of order at most 2^{n+2}. Moreover $(h+1)h = h - 1$, so $\pm h - 1 \in (h+1)\langle 2 \rangle$ and $\langle m \rangle \subseteq H_n$. The group $\langle m \rangle$ has the order 2^{n+2} of the Frobenius orbit of P_n, which is the degree of c_{n+1}. Hence $\langle m \rangle = H_n$, and $\varphi(r) \geq 2^{n+2}$ gives $r > 2^{n+2}$.

Now let $r = F_n = h^2 + 1$, and note that $\langle 2 \rangle = \{\pm 2^b : 0 \leq b < 2^n\}$ modulo F_n. Since $(h+1)2^{b+2^{n-1}} \equiv (h-1)2^b$, the residues of H_n in $(0, F_n)$ are 2^b and $F_n - 2^b$ for $0 \leq b < 2^n$, together with $(h \pm 1)2^b$ and $F_n - (h \pm 1)2^b$ for $0 \leq b < 2^{n-1}$. The odd ones are 1, $h \pm 1$, and residues $F_n - y$ with $y \leq (h+1)2^{2^{n-1}-1}$, which are at least $2^{2^{n-1}} + 1 - h/2 > h$. Since $-1 \in H_n$, the first claim follows.

For the second, suppose $k' \equiv \epsilon k 2^b$ or $k' \equiv \epsilon k(h+1)2^b$ with $\epsilon = \pm 1$ and $0 \leq b < 2^n$. In the first case, if $k2^b < F_n/2$, then $k' - \epsilon k 2^b$ is a nonzero integer of absolute value below F_n: it is odd when $b \geq 1$, and $k' \neq \pm k$ when $b = 0$. If $k2^b \geq F_n/2$, multiplication by $2^{2^n - b} < 2k$ gives $2^{2^n - b}k' + \epsilon k \equiv 0$, an odd integer of absolute value below $2kk' + k < F_n$. In the second case, multiplication by $h - 1$ and $(h+1)(h-1) \equiv -2$ give $k'(h-1) + \epsilon k 2^B \equiv 0$ with $B = b + 1$. If $k2^B < F_n/2$, the left side is an odd integer of absolute value below $k'h + F_n/2 < F_n$. Otherwise multiplication by $2^{2^n - B} < 2k$ gives $k'(h-1)2^{2^n - B} - \epsilon k \equiv 0$, where the left side has absolute value below $2kk'(h-1) + k < F_n$ and is nonzero: by parity if $B < 2^n$, and because $k'(h-1) \geq h - 1 > k$ if $B = 2^n$. Each case contradicts F_n dividing a nonzero integer of smaller absolute value. □

For $2 \leq n \leq 6$ a direct computation gives $r = F_n$; whether $\mathrm{ord}(P_n) = F_n$ for every n is itself a selected-order question. When $r = F_n$ and $n \geq 3$, Theorem 4.1 of [21] does not apply to $g(P_n)$, because $F_n > 2^{3(n+2)/2}$. Under $r = F_n$ the usual odd-multiplier argument also meets a counting obstruction. Its standard degree estimate separates products when the sum of their function degrees, proportional to $\sum k^2$, is below the degree $d = 2^{n+2}$ of P_n. For $n \geq 4$, any two distinct odd multipliers within this budget satisfy $kk' < d \leq h/2$, so each Frobenius coset contains at most one, and this sufficient budget cannot supply the growing family of multipliers that Voloch's argument uses. The first nontrivial odd multiplication map representing Frobenius itself, $[h - 1]$, has degree $(h-1)^2 > 2^{2^n - 1}$. None of this excludes sharper degree estimates that exploit cancellation, even multipliers, other counting schemes, non-integer endomorphisms, or other realizations of u_n. A realization satisfying Voloch's hypotheses would be an independent ancestry

input in the sense of Section 8; none is known. Popovych's effective exclusions [14, Theorem 2], valid for $i \geq 2$, use trivial intersections between the cyclic groups of parent coordinates, such as $\langle x_{i-1}\rangle \cap \langle x_{i-2}\rangle = 1$. For the trace-compatible Conway parents that intersection is nontrivial when $i \geq 3$: by Proposition 4.14(a), $a_{i-2}^3 = N(a_{i-1}) = a_{i-1}^{F_{i-1}}$ lies in $\langle a_{i-1}\rangle \cap \langle a_{i-2}\rangle$ and is not 1.

5 The ordinary arm: a marked cyclotomic value

Fix an odd prime r, write $b = b_r$, and abbreviate $x_a = x_{r,a}$, $y_a = x_a + 1$, and $F_a = F_{r,a}$.

Proposition 5.1 (projective order criterion)**.** *For $a \geq 1$,*

$$\mathrm{N}_{F_a/F_{a-1}}(y_a) = y_{a-1}, \qquad |F_a^\times / F_{a-1}^\times| = L_{r,a}.$$

If $p \neq r$ divides $L_{r,a}$, then the corresponding singleton row has

$$m_p = 1 \quad \Longleftrightarrow \quad v_p\big(\mathrm{ord}(y_a F_{a-1}^\times)\big) = v_p(L_{r,a}).$$

Every singleton row with component r^a arises this way.

Proof. The relative conjugates of x_a are ξx_a for $\xi \in \mu_r$; evaluating their binomial at one gives the norm identity. The quotient order is the quotient of the two finite-field unit orders. The cyclotomic divisor criterion gives $\mathrm{ord}_p(2^b) = r^a$ for $p \neq r$ dividing $L_{r,a}$, and Lenstra's component theorem identifies precisely the singleton rows. The lower unit group has no p-part, so Proposition 2.1 gives the last equivalence. □

The full recursive ancestry can also be retained in one binary polynomial. Let $P_r \in \mathbb{F}_2[T]$ be the minimal polynomial of $x_0 = \alpha_r$.

Proposition 5.2 (bottom-ancestor divisibility)**.** *Put $q = |F_{a-1}|$, $N = L_{r,a}$, and let $p \neq r$ divide N. Then $\gcd(p, q-1) = 1$. Write $d = N/p$, and choose $e, t \geq 0$ such that*

$$pe = 1 + t(q-1).$$

Then the current p-coordinate of $y_a F_{a-1}^\times$ is deficient if and only if

$$P_r(T^{r^a}) \mid (1+T)^d + (1+T^r)^e \qquad \text{in } \mathbb{F}_2[T].$$

Proof. Indeed, $N \equiv r \pmod{q-1}$, so $\gcd(N, q-1) = r$. The polynomial $P_r(T^{r^a})$ is the minimal polynomial of x_a. The coordinate is deficient exactly when $c = (1+x_a)^d$ lies in $F_{a-1}^\times$. In that case the norm identity gives

$$c^p = (1+x_a)^N = 1 + x_a^r.$$

Since $c^{q-1} = 1$ and $pe = 1 + t(q-1)$, $c = (c^p)^e = (1+x_a^r)^e$. Conversely this equality puts c in F_{a-1}. In characteristic two the equality is precisely the vanishing at x_a of the displayed binary polynomial, which is equivalent to divisibility by its minimal polynomial. □

This is a lossless selector-retaining reduction, not a proof of arm O: the remaining assertion is that the displayed divisibility never occurs for the literal polynomial P_r and every current factor p.

For a polynomial $B(T) = \sum_m b_m T^m$ and $0 \leq j < n$, define its j-th n-section by

$$\mathrm{Sec}_{n,j}(B)(U) = \sum_{k \geq 0} b_{j+kn} U^k.$$

Lemma 5.3 (polynomial section criterion)**.** *Let k be a field, let $0 \neq P \in k[U]$, and let $n \geq 1$. Then*

$$P(T^n) \mid B(T) \quad \Longleftrightarrow \quad P(U) \mid \mathrm{Sec}_{n,j}(B)(U) \quad (0 \leq j < n).$$

Proof. Every polynomial has the unique decomposition $B(T) = \sum_{j<n} T^j \,\mathrm{Sec}_{n,j}(B)(T^n)$. Multiplication by $P(T^n)$ preserves each residue class of exponents modulo n. Applying the decomposition to the quotient proves the forward implication, and summing the sectionwise quotients proves the converse. □

Corollary 5.4 (Lucas-section ordinary case)**.** *Put $n = r^a$, let $p \neq r$ divide $N = L_{r,a}$, and put $d = N/p$. If some $0 \leq j < n$ satisfies*

$$r \nmid j, \qquad \binom{d}{j} \equiv 1 \pmod 2, \qquad d - j < bn,$$

then the current p-coordinate of $y_a F_{a-1}^{\times}$ has full primary order. In particular this holds if $d \leq bn$, and hence when N/r is a prime distinct from r.

Proof. In the polynomial of Proposition 5.2, the j-th section receives no term from $(1+T^r)^e$, since $r \mid n$ but $r \nmid j$. Its other summand has constant coefficient one and degree at most $\lfloor (d-j)/n \rfloor < b$. It is therefore a nonzero polynomial of degree less than $\deg P_r = b$, so Lemma 5.3 rules out the required divisibility. Finally, d is odd, so when $d \leq bn$ one may take $j = 1$. If $p = N/r$, then $d = r \leq br^a$. □

There is a selector-sensitive strengthening obtained by retaining the two opposite nonzero sections at the top Kummer step. For a monic polynomial H with constant coefficient one, write

$$H^*(U) = U^{\deg H} H(U^{-1})$$

for its reciprocal.

Proposition 5.5 (reciprocal doubling in the ordinary arm)**.** *Put $n = r^a$, let $p \neq r$ divide $N = L_{r,a}$, and put $d = N/p$. For $0 \leq j < r$ define the top r-sections*

$$A_j(W) = \mathrm{Sec}_{r,j}((1+T)^d)(W) = \sum_{k \geq 0} \binom{d}{j+kr} W^k.$$

Set $s = r^{a-1}$ and

$$M(W) = P_r(W^s), \qquad M^*(W) = P_r^*(W^s).$$

Then $r \mid d$, and, writing $h = d/r$, one has the exact section reciprocity

$$A_{r-1}(W) = W^{h-1} A_1(W^{-1}).$$

If the current p-coordinate is deficient, then both M and M^* divide A_1. Consequently, if $P_r \neq P_r^*$, deficiency forces

$$d \geq 2br^a + r.$$

Equivalently, when $P_r \neq P_r^*$, every current factor satisfying $d \leq 2br^a$ has full primary order.

This non-self-reciprocity condition is independent of the tower level. More explicitly, if $f = \mathrm{ord}_r(2)$, then $f \mid b$, and it holds automatically whenever f is odd or b/f is even.

Proof. The relative Kummer construction has $x_a^r = x_{a-1}$ and degree r; as above, its nonpower radicand gives $r \mid 2^b - 1$. Therefore

$$N = \sum_{i=0}^{r-1} (2^b)^{ir^{a-1}} \equiv r \equiv 0 \pmod{r}.$$

As $p \neq r$, this gives $r \mid d$.

Put $K = F_{a-1}$, $z = x_a$, and $w = z^r = x_{a-1}$. The polynomial M is the minimal polynomial of w over $\mathbb{F}_2$. If the coordinate is deficient, then $(1+z)^d \in K$. Expanding in the K-basis $1, z, \ldots, z^{r-1}$ shows that

$$A_j(w) = 0 \qquad (1 \leq j < r),$$

so M divides every nonzero top section. Since $r \mid d$, also $z^d = w^{d/r} \in K$, and therefore

$$(1 + z^{-1})^d = z^{-d}(1+z)^d \in K.$$

Expansion in the inverse basis gives $A_j(w^{-1}) = 0$ for $1 \leq j < r$. The minimal polynomial of w^{-1} is M^*, so M^* divides A_1.

The displayed section reciprocity follows directly from $\binom{d}{\ell} = \binom{d}{d-\ell}$ and $d = rh$: the coefficient indexed by $1 + kr$ is paired with the one indexed by $(r-1) + (h-k-1)r$. The polynomial P_r has constant coefficient one because $x_0 \neq 0$, and hence M and M^* are monic irreducibles. If $P_r \neq P_r^*$, then

$$M^*(W) = P_r^*(W^s) \neq P_r(W^s) = M(W),$$

so they are coprime. Thus $MM^* \mid A_1$ under deficiency. But A_1 is nonzero, since its constant coefficient is $\binom{d}{1} = 1$, and

$$\deg A_1 \leq h - 1 = \frac{d}{r} - 1, \qquad \deg(MM^*) = 2bs.$$

It follows that $d/r - 1 \geq 2br^{a-1}$, which is the claimed lower bound. Because $r \mid d$, its contrapositive is equivalently $d \leq 2br^a$.

It remains to prove the arithmetic sufficient condition. The first Kummer step shows that x_0 is not an r-th power in $F_0^\times$: otherwise, since $\mu_r \subset F_0$, the chosen r-th root x_1 would already lie in F_0. Hence $r \mid \operatorname{ord}(x_0)$. If $P_r = P_r^*$, then $x_0^{-1} = x_0^{2^j}$ for some $0 < j < b$. Since the Frobenius orbit has length b, this forces b even, $j = b/2$, and

$$2^{b/2} \equiv -1 \pmod{\operatorname{ord}(x_0)}, \quad \text{hence} \quad 2^{b/2} \equiv -1 \pmod{r}.$$

Writing $f = \operatorname{ord}_r(2)$, the last congruence requires f even and b/f odd. The stated alternatives are the contrapositive. □

The apparent self-reciprocal exception in Proposition 5.5 cannot be the first failure of the 0/1/4 rule. The algebraic input is the following affine Kummer lemma.

Lemma 5.6 (affine odd-Kummer nonunitarity)**.** *Let E be a field of characteristic two, let $q > 1$ be odd, and let $a, u \in E^\times$. Suppose that t has minimal polynomial $X^q - a$ over E. Then, for every $k \geq 0$,*

$$(t+u)^{2^k+1} \neq 1.$$

Proof. Put $Q = 2^k$ and write $Q = Aq + v$ with $0 < v < q$; the residue is nonzero because q is odd. If the displayed equality held, then Frobenius additivity would give

$$0 = (t+u)^{Q+1} + 1 = t^{Q+1} + ut^Q + u^Q t + u^{Q+1} + 1. \tag{$*$}$$

Reduce this expression in the E-basis $1, t, \ldots, t^{q-1}$ using $t^q = a$. If $v = 1$, the coefficient of t^2 is $a^A \neq 0$. If $2 \leq v \leq q-2$, the coefficient of t^{v+1} is $a^A \neq 0$. If $v = q-1$, the coefficient of t^{q-1} is $ua^A \neq 0$. Each contradicts $(*)$, and these cases exhaust the possible nonzero residues. □

Proposition 5.7 (bottom reciprocity cannot be a first failure)**.** *Let $r \neq 3$ be an odd prime. If the 0/1/4 rule holds at the row r itself, then the literal bottom polynomial is not self-reciprocal:*

$$P_r \neq P_r^*.$$

The excluded prime really is exceptional: $\alpha_3 = 2$ and $P_3(X) = X^2 + X + 1 = P_3^*(X)$.

Proof. Write $f = \mathrm{ord}_r(2)$ and use the Conway–Lenstra component decomposition

$$\alpha_r = \kappa_f + m_r, \qquad \kappa_f = \sum_{q \in \mathcal{Q}(f)} \kappa_q.$$

If $P_r = P_r^*$ and $b = d(\alpha_r)$, then b is even and

$$\alpha_r^{2^{b/2}+1} = 1. \tag{$**$}$$

Suppose first that $\mathcal{Q}(f)$ contains more than one component. The rule gives $m_r = 0$. Choose the largest odd component s^c and put $t = \kappa_{s^c}$. Let E be generated by α_s, all the other-prime components, and the finite translate, all at once. The element α_s lies strictly below the s-component, and every other component prime is smaller than s by the choice of the largest odd component. Thus none of these generators contributes any s-primary field degree. The full s^c-Kummer component therefore remains after passing to E, and the component theorem gives

$$t^{s^c} = \alpha_s, \qquad [E(t) : E] = s^c;$$

in particular $1, t, \ldots, t^{s^c-1}$ is a power basis. The sum u of the remaining components is a nonzero element of E, by uniqueness of the Conway component normal form. Lemma 5.6, with $Q = 2^{b/2}$, contradicts $(**)$.

If $\mathcal{Q}(f) = \{s^c\}$ is a singleton odd prime power, the rule gives $m_r = 1$, except when $f = 2 \cdot 3^j$, when it gives $m_r = 4$. In the first case take $t = \kappa_{s^c}$ and $u = 1$. In the exceptional case take $t = \kappa_{3^j}$ and $u = 4$; adjoining the finite field containing 4 to the coefficient field leaves the odd relative degree 3^j. The same lemma again contradicts $(**)$.

The remaining possibility is a singleton even component. Then the rule gives $m_r = 0$, and $f = 2^j$. The selected Artin–Schreier generator $c = \kappa_{2^j}$ has degree 2^j and its nontrivial conjugate over the half field is $c + 1$. Since $r \neq 3$, one has $2^j > 2$. Equation $(**)$ would give $c(c+1) = 1$, so c would satisfy $X^2 + X + 1$, contradicting its degree. This proves the assertion. For $r = 3$, the displayed polynomial is the defining quadratic and is self-reciprocal. □

Corollary 5.8 (minimal-counterexample reciprocal sieve)**.** *Suppose that p is the least odd prime at which the 0/1/4 rule fails and that its putative failure lies in arm O at (r, a, p). Then $P_r \neq P_r^*$. Consequently, with $d = L_{r,a}/p$, a failure must satisfy*

$$d \geq 2b_r r^a + r;$$

thus no least counterexample in arm O can have $d \leq 2b_r r^a$.

Proof. The cyclotomic divisor criterion gives $\mathrm{ord}_p(2^{b_r}) = r^a$, so $r \mid p-1$ and hence $r < p$. The rule therefore holds at r by minimality. Apply Proposition 5.7 and then Proposition 5.5. □

The full Kummer orbit gives a second sieve which is substantially stronger when the tower level is large.

Proposition 5.9 (Kummer-ratio packing over minimal coefficient fields)**.** *Let r be any odd prime, $b = b_r$, and $a \geq 1$, and let o be the order of the literal selected class $(1+x_{r,a})F^{\times}_{r,a-1}$. For $1 \leq m \leq a$ put*

$$t_m = \mathrm{ord}_{r^m}(2), \qquad D_m = r^{m-1}(r-1), \qquad R_m = \frac{br^a}{t_m}.$$

These are positive integers, and $R_m \geq r$. Write

$$\mathcal{R}(D,s) = \sum_{t=0}^{s} 2^t \binom{D}{t}\binom{s}{t}$$

and define the parity-sensitive bound

$$\mathcal{B}_r(D,R) = \begin{cases} \mathcal{R}(D,R/2), & R \text{ even}, \\ \mathcal{R}(D,(R-1)/2) + \left\lceil \dfrac{\mathcal{R}(D,(R+1)/2) - \mathcal{R}(D,(R-1)/2)}{r} \right\rceil, & R \text{ odd}. \end{cases}$$

Then

$$B_{r,a} := \max_{1 \leq m \leq a} \mathcal{B}_r(D_m, R_m) \leq o.$$

Consequently, a deficient current prime $p \neq r$ must satisfy $L_{r,a}/p \geq B_{r,a}$.

Proof. Put $v = v_r(2^b - 1) \geq 1$, $K = F_{r,a-1}$, $L = F_{r,a}$, $x = x_{r,a}$, and $y = 1 + x$. Fix m. LTE gives

$$v_r(2^{br^{a-1}} - 1) = v + a - 1 \geq a \geq m.$$

Thus $t_m \mid br^{a-1}$ and the smallest finite field containing μ_{r^m} is

$$H_m = \mathbb{F}_{2^{t_m}} \subset K.$$

Since x generates L over $\mathbb{F}_2$, its minimal polynomial over H_m has degree $[L : H_m] = R_m \geq r$ and nonzero constant coefficient.

Let σ be the $|K|$-Frobenius on L, and put $u = \sigma(y)/y$. The Hilbert–90 isomorphism $L^{\times}/K^{\times} \longrightarrow \ker \mathrm{N}_{L/K}$ gives $\mathrm{ord}(u) = o$: its kernel is $K^{\times}$ and both finite groups have order $L_{r,a}$. Choose representatives $\mathcal{C}_m$ for μ_{r^m}/μ_r. For $c \in \mathcal{C}_m$ and $\zeta \in \mu_r \setminus \{1\}$ set

$$u_{c,\zeta} = \frac{1 + \zeta cx}{1 + cx}.$$

There are exactly D_m such ratios, all with coefficients in H_m. Their numerators and denominators are nonzero: otherwise $x \in H_m$, contrary to $R_m > 1$.

Every ratio belongs to $\langle u \rangle$. Indeed, $m \leq a$ makes cx a root of the literal irreducible polynomial $P_r(T^{r^a})$, so $cx = x^{2^e}$ for some e. The automorphism σ fixes c, and $\sigma(x)/x$ generates μ_r. Choose k such that $\sigma^k(cx) = \zeta cx$. Commutativity of absolute and relative Frobenius gives

$$u_{c,\zeta} = u^{2^e(1+|K|+\cdots+|K|^{k-1})}.$$

Index the ratios by I, with $|I| = D_m$, and evaluate integer vectors by $U_w = \prod_{i\in I} u_i^{w_i}$. A signed ball $\sum_i |w_i| \le s$ has exactly $\mathcal{R}(D_m, s)$ elements: choose its t nonzero coordinates, their signs, and their positive absolute values with sum at most s.

Suppose first that R_m is even. Include the entire signed ball of radius $R_m/2$. If $U_w = U_{w'}$, put $g = w - w'$ and clear denominators without cancelling common factors. This gives $A(x) = B(x)$ with $A, B \in H_m[T]$, both constant coefficients one, and both degrees $\sum_i |g_i| \le R_m$. The minimal polynomial of x divides $A - B$. A nonzero difference would therefore be a scalar multiple of that minimal polynomial, contradicting its zero constant coefficient. Hence $A = B$.

If R_m is odd, put $s = (R_m - 1)/2$ and include the inner ball of radius s together with a largest fibre, in the shell of radius $s + 1$, of the leading-phase character

$$\chi(w) = \prod_{(c,\zeta)} \zeta^{w_{c,\zeta}} \in \mu_r.$$

The chosen family has cardinality at least $\mathcal{B}_r(D_m, R_m)$. For a collision with at least one inner-ball vector, the same cross-product degree is at most $2s + 1 = R_m$. For two shell vectors, it is at most $2s + 2 = R_m + 1$, but equality of leading phases cancels the leading coefficient. Thus $\deg(A - B) \le R_m$ again, and the same zero-constant argument gives $A = B$.

In either parity, different c-cosets have disjoint stars of linear factors. Within each star, valuation at $1 + \zeta cT$ recovers the corresponding signed exponent. Unique factorization makes $A = B$ force $g = 0$. The chosen vectors therefore inject into the same cyclic group $\langle u\rangle$. The normalized rational evaluation injection and the resulting bound by $\mathrm{ord}(u)$ are kernel-checked in `ExcessOrdinaryPacking`; finite-field ancestry, the disjoint-factor comparison, and the lattice count remain paper-level inputs. Taking the maximum over m proves the bound. Under a deficient current p-coordinate, $o \mid L_{r,a}/p$, proving the stated sieve. □

Corollary 5.10 (intermediate-field recovery and explicit growth)**.** *Put $v = v_r(2^b - 1)$. For $0 \le j < a$, set*

$$m_j = \min\{a, j + v\}, \quad D'_j = r^{m_j - 1}(r - 1), \quad R'_j = r^{a-j}, \quad s_j = (R'_j - 1)/2.$$

Then $B_{r,a}$ is at least

$$\mathcal{R}(D'_j, s_j) + \left\lceil \frac{\mathcal{R}(D'_j, s_j + 1) - \mathcal{R}(D'_j, s_j)}{r} \right\rceil$$

for every j. The case $j = a - 1$ uses $D = r^{a-1}(r - 1)$ and signed radius $(r - 1)/2$. Moreover,

$$\log_2 o \ge (\log_2 3)(1 - 1/r)\sqrt{r^a} - \tfrac{1}{2}\log_2 3.$$

Proof. LTE puts $\mu_{r^{m_j}}$ in $F_{r,j}$, so $t_{m_j} \mid br^j$ and hence $R_{m_j} \ge R'_j$. If these degrees are equal, they are odd and the displayed term is exactly $\mathcal{B}_r(D_{m_j}, R_{m_j})$. If they are unequal, $R_{m_j} \ge R'_j + 1 = 2(s_j + 1)$, so the larger degree bound contains the entire ball of radius $s_j + 1$, which already dominates the displayed inner-ball-plus-partial-shell term.

For the growth assertion choose $j = \lfloor (a - 1)/2 \rfloor$ and put $h = \min\{D'_j, s_j\}$. The inner ball contains $\{-1, 0, 1\}^h$, so its size is at least 3^h. Since $v \ge 1$, $D'_j \ge (r - 1)r^j$. If $a = 2k$, then $h \ge (r - 1)r^{k-1}$. If $a = 2k + 1$, then $h \ge (r^{k+1} - 1)/2$. In both cases $h \ge (1 - 1/r)r^{a/2} - 1/2$, already for $r \ge 3$. □

For fixed r and bottom ancestry this packing bound is exponential in $\sqrt{r^a}$, whereas $L_{r,a}$ is exponential in r^a. To see the upper scale as well, put $v_0 = v_r(2^{t_1} - 1)$. Order lifting gives $t_m = t_1 r^{\max(0,m-v_0)}$, so $D_m R_m = O(r^a)$ uniformly in m. The elementary estimate

$$\mathcal{R}(D,s) \le \sum_{t\ge 0} \frac{(2Ds)^t}{(t!)^2} \le \exp(2\sqrt{2Ds})$$

and $\mathcal{B}_r(D,R) \le \mathcal{R}(D,\lceil R/2\rceil)$ therefore give $\log B_{r,a} = O(\sqrt{r^a})$. Thus the theorem proves a selected large-factor range, not universal arm O. Bounds from different coefficient fields cannot simply be multiplied: they count subsets of the same cyclic group, and a mixed collision must be tested over the largest coefficient field present. The statement includes $r = 3$; its cubic interpretation is given in Proposition 6.4.

Corollary 5.11 (minimal-counterexample congruence-gap regime)**.** *Suppose that p is the least odd prime at which the $0/1/4$ rule fails and that its putative failure lies in arm O at (r,a,p), so $r \ne 3$. Let $N = L_{r,a}$, let o be the selected projective order, write $N = o\Delta$, and let $\Delta^{(r')}$ be the prime-to-r part of Δ. Let $B_{r,a}$ be the minimal-coefficient-field maximum in Proposition 5.9, and put*

$$M = 2r^a, \qquad A_{<p} = \prod_{\substack{q<p,\ q\ne r\\ q\mid N}} q^{v_q(N)}, \qquad B_* = \max\{B_{r,a}, A_{<p}\}.$$

Then every prime divisor of $\Delta^{(r')}$ is at least p and is congruent to 1 modulo M, and

$$N < B_* p^2 \quad\Longrightarrow\quad \Delta^{(r')} = p.$$

In the larger range

$$N < B_* p(p+M) \quad\Longrightarrow\quad \Delta^{(r')} = p \quad or \quad p^2 \mid \Delta^{(r')}.$$

If $v_p(N) = 1$, the second alternative is impossible.

Proof. The deficient p-coordinate gives $p \mid \Delta^{(r')}$. If a prime $q < p$, $q \ne r$, divided Δ, then $q \mid L_{r,a}$ would be another eligible arm-O row with deficient q-valuation, contradicting the minimality of p. Thus $\Delta^{(r')}$ is p-rough. For every such prime $q \ne r$, the cyclotomic divisor theorem gives $\mathrm{ord}_q(2^{b_r}) = r^a$. Since q is odd and r^a is odd, this implies $q \equiv 1 \pmod{2r^a}$.

The unconditional part of Proposition 5.9 gives $B_{r,a} \le o$. Minimality also forces the full q-primary part of N into o for every $q < p$, $q \ne r$, so $A_{<p} \mid o$ and therefore $B_* \le o$. Hence

$$B_* \Delta^{(r')} \le B_* \Delta \le o\Delta = N.$$

If $\Delta^{(r')} \ne p$, its quotient by p has a prime divisor at least p, proving the first implication. If $p^2 \nmid \Delta^{(r')}$, that prime is distinct from p and congruence spacing makes it at least $p+M$; hence $\Delta^{(r')} \ge p(p+M)$. This proves the second implication and the valuation-one conclusion. A surviving square factor similarly makes p a base-2^{b_r} Wieferich prime, since $r^a \mid p-1$ and $(2^{b_r})^{r^a} \equiv 1 \pmod{p^2}$. □

Let $N_a = \mathrm{ord}(x_a)$. The Kummer relation $x_a^r = x_{a-1}$ and the relative degree r give $N_a = rN_{a-1}$. Choose the roots and primes recursively: ξ_{N_a} is a primitive complex N_a-th root satisfying

$$\xi_{N_a}^r = \xi_{N_{a-1}},$$

and $\mathfrak{P}_a \mid 2$ is the prime whose residue sends ξ_{N_a} to x_a. Since $x_a^r = x_{a-1}$, this makes $\mathfrak{P}_a$ restrict to $\mathfrak{P}_{a-1}$. Put $\varepsilon_a = 1 - \xi_{N_a}$. For a row prime p, use the intrinsic residue-field notation

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a}\right)_p := \overline{\varepsilon_a}^{(|F_a|-1)/p} \in \mu_p(F_a).$$

This is an Euler symbol in the selected residue field. It agrees with the usual local power-residue/Artin symbol after adjoining μ_p and choosing the prime above $\mathfrak{P}_a$.

Theorem 5.12 (selected cyclotomic-value form)**.** *For $p \neq r$ dividing $L_{r,a}$, arm O at (r, a, p) is equivalent to*

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a}\right)_p \neq 1.$$

The compatible primes $\mathfrak{P}_a$ form a unique inert ray above two: $\mathfrak{P}_a$ is the unique prime above $\mathfrak{P}_{a-1}$, with relative residue degree r, and

$$\mathrm{N}_{\mathbb{Q}(\xi_{N_a})/\mathbb{Q}(\xi_{N_{a-1}})}(1 - \xi_{N_a}) = 1 - \xi_{N_{a-1}}.$$

Nevertheless, residue-field corestriction kills the new p-coordinate and does not determine the displayed residue symbol.

Proof. The reduction of ε_a is y_a. The unramified power-residue formula gives

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a}\right)_p = y_a^{(|F_a|-1)/p},$$

so Proposition 2.1 and Proposition 5.1 give the equivalence.

The Kummer steps increase the r-primary part of N_a by one and give $\mathrm{ord}_{N_a}(2) = br^a$. The relative cyclotomic degree and the residue-degree ratio are therefore both r, forcing one prime above the preceding selected prime. The global norm identity is the product over the r-th roots of unity. Finally, $p \mid \Phi_{r^a}(2^b)$ but $p \nmid 2^{br^{a-1}} - 1$, so the lower residue field has trivial p-Kummer quotient. Its norm receives no information about the new selected class. □

The word *value* in the theorem is deliberate. The element $1 - \xi_{N_a}$ is a cyclotomic number and is an S-unit for the primes dividing N_a; it is a global unit exactly when N_a is not a prime power. For example, in the $r = 7$ chain one has $N_a = 7^{a+1}$ and its absolute norm is 7. This distinction does not alter reduction modulo the selected prime over two or the relative norm identity, but any global Kummer formulation must retain the conductor primes in S (or in the ray modulus).

The selected symbol has an exact Jacobi–Fourier expansion. Let $Q = |F_a|$, let $H = \langle x_a \rangle \subset F_a^\times$ have order N_a, put $M = (Q-1)/N_a$, and let χ be a complex-valued order-p power character. Extend complex-valued multiplicative characters by zero at zero and define

$$f(h) = \chi(1-h), \qquad \widehat{f}(\theta) = \sum_{h \in H} f(h)\theta(h)^{-1} \quad (\theta \in \widehat{H}).$$

For an extension ψ_θ of θ to $F_a^\times$, and with $H^\perp$ the character group of $F_a^\times/H$, one has

$$\widehat{f}(\theta) = \frac{1}{M} \sum_{\lambda \in H^\perp} J(\psi_\theta^{-1}\lambda, \chi), \qquad f(x_a) = \frac{1}{N_a} \sum_{\theta \in \widehat{H}} \theta(x_a)\widehat{f}(\theta).$$

Indeed, the character average $M^{-1}\sum_{\lambda \in H^\perp} \lambda$ is the indicator of H; inserting it in the first Fourier sum gives the Jacobi sums, and Fourier inversion gives the second identity. This is a fully explicit

reciprocity formula, but the selected label survives in $\theta(x_a)$ and the Jacobi unit phases. Ideal factorization or valuation formulas for the Jacobi sums do not by themselves determine this inverse-DFT entry. This separation of the ideal and p-adic unit conditions is also visible in Uehara's cyclotomic-unit criterion [19]; it supplies an exact test, not a uniform nonvanishing theorem for the Conway-selected entry.

After adjoining μ_p and retaining the conductor primes in the S-ray modulus, this residue-field condition becomes a marked Artin-symbol assertion, not merely a statement about decomposition or conductor. Its ray character splits into a ramified local logarithm and an unramified ideal-class character evaluated at $[\mathfrak{P}_a]$. Stickelberger annihilation reduces the latter to an explicit decomposition-invariant quotient, but does not make it vanish uniformly. This is a genuine boundary: at the actual Conway conductor $r = 11$, $a = 1$, $p = 23$, one contributing Stickelberger scalar vanishes with exact 23-adic valuation one. The corresponding unramified degree-23 class line is nontrivial, and every prime over two has nontrivial Artin symbol on it, although the literal row still has $m_{23} = 1$. Thus successful rows need not have a zero unramified term; the open assertion is noncancellation in the actual selected sum.

6 The cubic arm: a selected Singer generator

Let $h = 3^k$, $\zeta_k = \kappa_{3^k}$, and $\gamma_k = \zeta_k + \zeta_k^{-1}$.

Proposition 6.1 (half-angle reduction)**.** *The element ζ_k is a primitive 3^{k+1}-st root of unity,*

$$\mathbb{F}_2(\zeta_k) = \mathbb{F}_{2^{2h}}, \qquad \mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}.$$

Moreover,

$$\operatorname{ord}(\zeta_k + 1) = 3^{k+1} \operatorname{ord}(\gamma_k),$$

and the translates by 2 and 3 have the same order. Hence the rows with $f(p) = 3^k$ have excess one precisely when γ_k has the full relevant primary order; over all levels this is equivalent to primitivity of every γ_k.

Proof. The defining Kummer equation gives $\zeta_k^{3^k} = \kappa_2$, a primitive cube root. The order of two modulo 3^{k+1} is $2h$, and the least exponent giving ± 1 is h, proving the field degrees. If s is the inverse of two modulo 3^{k+1}, then

$$\zeta_k + 1 = \zeta_k^s(\zeta_k^s + \zeta_k^{-s}), \qquad (\zeta_k^s + \zeta_k^{-s})^2 = \gamma_k.$$

The factors lie in the coprime circle and real parts of the unit group, giving the order identity. Frobenius gives the other translates, and Proposition 2.1 finishes the reduction. □

The half-angle test also has a lossless group-algebra form that removes all denominators. Put

$$J_h(X) = X^{2h} + X^h + 1, \qquad B_d(X) = (X^2 + 1)^d + X^d \in \mathbb{F}_2[X].$$

Since J_h is the minimal polynomial of ζ_k and $\gamma_k = \zeta_k^{-1}(\zeta_k^2 + 1)$, one has

$$\gamma_k^d = 1 \quad \Longleftrightarrow \quad J_h(X) \mid B_d(X). \tag{1}$$

Equivalently, in $\mathbb{F}_2[X, X^{-1}]/(X^{3h} - 1)$ the parity vector of $X^d(1 + X^{-2})^d$ belongs to the affine coset $1 + (J_h)$, not to the ideal (J_h) itself. Thus arm C is exactly the universal assertion that, for every prime $\ell \mid 2^h - 1$,

$$J_h(X) \nmid B_{(2^h-1)/\ell}(X).$$

Lucas's theorem makes the support of B_d explicit from the binary digits of d, but it does not by itself prove these selected nondivisibilities. The denominator-free equivalence above is also kernel-checked as `halfAngle_pow_eq_one_iff_minpoly_dvd`; it is a reduction of C, not a proof of C.

There is also an exact stable-polynomial formulation. Put $P_0(X) = X$ and

$$P_{j+1}(X) = P_j(X)^3 + P_j(X)^2 + 1,$$

so that $P_j = f^{\circ j}$ for $f(X) = X^3 + X^2 + 1$. In characteristic two, $t(X) = X + 1$ conjugates $g(X) = X^3 + X$ to f: $f = t \circ g \circ t$. The triple-angle identity gives $g(\gamma_j) = \gamma_{j-1}$ with $\gamma_0 = 1$, hence $b_j = \gamma_j + 1$ is a root of P_j. Since b_j has degree 3^j, P_j is its minimal polynomial. Lin and Wang prove that f is stable, equivalently that all these iterates are irreducible [12]; their theorem does not assert multiplicative primitivity.

Indeed, this missing strengthening is exactly arm C. With $q = 2^{3^{j-1}}$, both γ_j and b_j have relative norm γ_{j-1} to $\mathbb{F}_q$, and

$$\gamma_j b_j^2 = \gamma_{j-1}.$$

Because $\gcd(q-1, q^2+q+1) = 1$, the norm propagates every old primary part. For a new prime $\ell \mid q^2+q+1$, the base-field factor γ_{j-1} is an ℓ-th power, while the displayed identity gives $[\gamma_j] + 2[b_j] = 0$ in the ℓ-Kummer quotient. Since ℓ is odd, γ_j is an ℓ-th power exactly when b_j is. Induction therefore shows that all P_j with $j \geq 1$ are primitive polynomials exactly when all γ_j with $j \geq 1$ are primitive. Stability supplies the degrees; primitive preservation is the original selected-order problem, not a known solution of it.

Put $q_k = 2^{3^{k-1}}$ and

$$U_k = \ker\left(\mathrm{N}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}\right), \qquad \eta_k = \gamma_k^{q_k - 1}, \qquad \eta_0 = 0.$$

Theorem 6.2 (selected norm-one coordinate)**.** *One has $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$. The element η_k generates $\mathbb{F}_{2^{3^k}}$ and is the selected root of*

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1.$$

Furthermore,

$$C_k \iff C_{k-1} \text{ and } \mathrm{ord}(\eta_k) = |U_k|,$$

where C_j denotes primitivity of γ_j.

Proof. The definition puts η_k in the norm-one group. Frobenius calculation from ζ_k gives the displayed cubic; its coefficients are respectively the trace, second symmetric function, and norm. A base-field element of U_k has order dividing $\gcd(q_k - 1, 3) = 1$, so the nontrivial cubic is irreducible. Its quadratic coefficient recursively recovers η_{k-1}, proving the field degree. Finally, the finite-field unit group splits into its base-field and norm-one factors, whose orders are coprime. The norm propagates the old part of γ_k and η_k detects exactly the new factor. □

The recursion selects one Frobenius orbit inside a Singer difference set. Indeed, for

$$R(x) = \frac{x^3 + x + 1}{x^2 + x}, \qquad W(x) = \frac{x + \omega}{x + \omega^2},$$

one has

$$R(\eta_k) = \eta_{k-1}, \qquad W(R(x)) = W(x)^3, \qquad W(\eta_k) = \zeta_k^2.$$

Thus the complete lower ancestry, not merely degree or trace, marks the orbit. The displayed conjugacy exposes its cyclotomic phase but does not evaluate the multiplicative order of η_k.

Proposition 6.3 (selected Singer sieve)**.** *Let $N = |U_k|$ and $h = 3^k$. Then*

$$\operatorname{ord}(\eta_k) \geq h(h-1)+1.$$

If a prime $\ell \mid N$ is missing from the selected order at its full valuation, then

$$\frac{N}{\ell} \geq h(h-1)+1.$$

Hence every prime $\ell > N/(h(h-1)+1)$ occurs with full valuation.

Proof. The selected Frobenius orbit has h points in the Singer $(N, q_k+1, 1)$ difference set. Its $h(h-1)$ ordered off-diagonal quotients are distinct nonidentity elements of $\langle \eta_k \rangle$, giving the first bound. If the full ℓ-part is missing, the orbit lies in the subgroup of order N/ℓ, which must contain all these quotients. □

The sieve is polynomial in h, while N is exponential. It proves a genuine selected large-factor range but not arm C.

Proposition 6.4 (intermediate-field cubic ratio packing)**.** *Let $k \geq 1$, $h = 3^k$, $H = 3^{k-1}$, $q = 2^H$, and $N = q^2+q+1$. Let $x = \zeta_k$ be the selected primitive 3^{k+1}-st root, put $\gamma_k = x + x^{-1}$, and put $\eta_k = \gamma_k^{q-1}$. For $0 \leq j < k$ define*

$$D_j = 2 \cdot 3^j, \qquad R_j = 3^{k-j}, \qquad s_j = (R_j - 1)/2,$$

$$B_j = \mathcal{R}(D_j, s_j) + \left\lceil \frac{\mathcal{R}(D_j, s_j+1) - \mathcal{R}(D_j, s_j)}{3} \right\rceil, \qquad B_{3,k} = \max_{0 \leq j < k} B_j.$$

These are the $r = 3$, $b_3 = 2$ parameters of Proposition 5.9, with $m = j+1$. Then

$$\operatorname{ord}(\eta_k) \geq \left\lceil \frac{B_{3,k}}{3} \right\rceil.$$

In particular, if a prime $\ell \mid N$ is missing from the order of η_k at its full valuation, then

$$\frac{N}{\ell} \geq \left\lceil \frac{B_{3,k}}{3} \right\rceil.$$

Proof. Write $L = \mathbb{F}_{q^6} = \mathbb{F}_{4^{3^k}}$ and $K = \mathbb{F}_{q^2} = \mathbb{F}_{4^{3^{k-1}}}$. We first identify exactly the order o of $(1+x)K^\times$ in $L^\times/K^\times$:

$$o = 3 \operatorname{ord}(\eta_k).$$

Indeed, write $A = \operatorname{ord}(\gamma_k)$. The half-angle identity gives $\operatorname{ord}(1+x) = 3^{k+1}A$, with $A \mid q^3-1$ and $3 \nmid A$. Since H is odd, $\gcd(q^3-1, q^2-1) = q-1$, while LTE gives $v_3(q^2-1) = k$. Consequently

$$\gcd(3^{k+1}A, q^2-1) = 3^k \gcd(A, q-1),$$

and the order in the cyclic quotient is

$$o = \frac{3^{k+1}A}{\gcd(3^{k+1}A, q^2-1)} = 3\frac{A}{\gcd(A, q-1)} = 3 \operatorname{ord}(\gamma_k^{q-1}).$$

The coprime circle/real order splitting and transfer of a lower bound through this factor of three are kernel-checked in `ExcessCubicPacking`; the finite-field half-angle, valuation, and component-gcd inputs remain paper-level.

Since $\operatorname{ord}_{3^m}(2) = 2 \cdot 3^{m-1}$, the minimal coefficient fields in Proposition 5.9 are exactly $\mathbb{F}_{4^{3^{m-1}}}$. Its unconditional bound gives $o \geq B_{3,k}$, and the identity $o = 3 \operatorname{ord}(\eta_k)$ gives the first assertion. A deficient full ℓ-part implies $\operatorname{ord}(\eta_k) \mid N/\ell$, proving the second. □

Corollary 6.5 (cubic square-root exponential bound)**.** *As $k \to \infty$, with $h = 3^k$,*

$$\log_2 \operatorname{ord}(\eta_k) \geq \left(\log_2 3 + \frac{5}{6}\right)\sqrt{h} - O(\log h).$$

The coefficient $\log_2 3 + 5/6 > 2.4182$ improves the partition coefficient $\pi/(3\log 2)$ in Proposition 6.6.

Proof. It is enough to retain one summand $2^t\binom{D_j}{t}\binom{s_j}{t}$ of the signed ball. Put $u = \sqrt{h}$ and $t = \lfloor u/2 \rfloor$. If $k = 2m$ is even, take $j = m - 1$. Then

$$D_j = \frac{2}{3}u, \qquad s_j = \frac{3}{2}u - \frac{1}{2}.$$

Writing $H_2(z) = -z\log_2 z - (1-z)\log_2(1-z)$, the elementary Stirling estimate gives

$$\log_2\left(2^t\binom{D_j}{t}\binom{s_j}{t}\right) = \left[\frac{1}{2} + \frac{2}{3}H_2\left(\frac{3}{4}\right) + \frac{3}{2}H_2\left(\frac{1}{3}\right)\right]u - O(\log h) = \left(\log_2 3 + \frac{5}{6}\right)u - O(\log h).$$

If $k = 2m + 1$ is odd, take $j = m$. Then

$$D_j = \frac{2}{\sqrt{3}}u, \qquad s_j = \frac{\sqrt{3}}{2}u - \frac{1}{2},$$

giving the coefficient

$$c_{\mathrm{odd}} = \frac{1}{2} + \frac{2}{\sqrt{3}}H_2\left(\frac{\sqrt{3}}{4}\right) + \frac{\sqrt{3}}{2}H_2\left(\frac{1}{\sqrt{3}}\right).$$

This is at least the even coefficient, as can be seen without numerical comparison. The function $g(x) = e^x H_2(\frac{1}{2}e^{-x})$ is strictly concave on $x > -\log 2$: with $z = \frac{1}{2}e^{-x} \in (0,1)$,

$$g''(x) = \frac{e^x}{\log 2}\left(-\log(1-z) - \frac{z}{1-z}\right) < 0.$$

Thus $g(x) + g(-x)$ decreases for $0 < x < \log 2$. The even coefficient uses $x = \log(3/2)$, whereas the odd coefficient uses $x = \log(2/\sqrt{3})$, which is smaller. Finally, dividing the order bound by three costs only $\log_2 3$, which is absorbed into $O(\log h)$. □

The bound remains subexponential in h, while N is exponential in h. It therefore leaves the smaller current prime factors unresolved.

The cyclotomic selector also permits a complementary partition bound. This is the usual low-degree collision argument for Gaussian periods, specialized to the composite conductor 3^{k+1} and therefore retaining the literal selected element rather than replacing it by an arbitrary normal generator.

Proposition 6.6 (selected norm-one partition bound)**.** *Let $n = 3^{k+1} = 3h$. For an integer $m \geq 1$ put*

$$\mathcal{A}_m = \{1 \leq j \leq m : 3 \nmid j\}, \qquad t_m = |\mathcal{A}_m|, \qquad s_m = \sum_{j \in \mathcal{A}_m} j.$$

Let

$$\mathcal{U}_h = \{1 \leq j < n : 3 \nmid j\},$$

choose $0 \le z_j < 2h$ *with* $2^{z_j} \equiv j \pmod n$*, and, for* $E \subseteq \mathcal{U}_h$*, put*

$$Q_E = \sum_{j \in E} 2^{z_j}, \qquad s_E = \sum_{j \in E} j.$$

For distinct $E, F \subseteq \mathcal{U}_h$*, set*

$$A = E \setminus F, \quad B = F \setminus E, \quad W = s_A + s_B, \quad j_0 = \min(A \cup B).$$

Then $\eta_k^{Q_E} \neq \eta_k^{Q_F}$ *if either*

$$s_A \not\equiv s_B \pmod 3 \ \text{ and } \ W < h, \qquad \text{or} \qquad s_A \equiv s_B \pmod 3 \ \text{ and } \ W - 2j_0 < h.$$

In particular, weighted symmetric-difference sum below h *rules out every collision.*

Put

$$R_h = \frac{h-1}{2}, \qquad \mathcal{P}_h = \sum_{r=0}^{R_h} [X^r] \prod_{\substack{j \ge 1 \\ 3 \nmid j}} (1 + X^j).$$

Then

$$\operatorname{ord}(\eta_k) \ge \mathcal{P}_h, \qquad \log_2 \mathcal{P}_h = \left(\frac{\pi}{3 \log 2} + o(1) \right) \sqrt{h}.$$

If $s_m < h$*, then also*

$$\operatorname{ord}(\eta_k) \ge 2^{t_m}.$$

Consequently

$$\log_2 \operatorname{ord}(\eta_k) \ge \left(\frac{\pi}{3 \log 2} + o(1) \right) \sqrt{h},$$

and the elementary initial-segment subfamily alone gives

$$\log_2 \operatorname{ord}(\eta_k) \ge \sqrt{\frac{4h}{3}} - O(1).$$

If a prime $\ell \mid N = |U_k|$ *is absent from the order of* η_k *at its full valuation, then* $N/\ell \ge \mathcal{P}_h$ *and, whenever* $s_m < h$*, also* $N/\ell \ge 2^{t_m}$*.*

Proof. Put $w = \zeta_k^2$. The identity $W(\eta_k) = w$ gives

$$\eta_k = \frac{\omega + \omega^2 w}{1 + w}.$$

The order of two modulo n is $2h$, so the exponents z_j in the statement exist. They satisfy

$$\eta_k^{2^{z_j}} = \omega^j \frac{1 + (\omega w)^j}{1 + w^j}.$$

A collision $\eta_k^{Q_E} = \eta_k^{Q_F}$ first permits cancellation of all indices in $E \cap F$: the relevant numerator and denominator factors are nonzero at w, because both w and ωw are primitive n-th roots. Put $A = E \setminus F$, $B = F \setminus E$, and $W = s_A + s_B$ as in the statement. Clearing the remaining denominators makes w a root of the difference between

$$\omega^{s_A} \prod_{j \in A} (1 + (\omega X)^j) \prod_{j \in B} (1 + X^j) \quad \text{and} \quad \omega^{s_B} \prod_{j \in B} (1 + (\omega X)^j) \prod_{j \in A} (1 + X^j).$$

Each polynomial has degree W. The element w is a primitive n-th root and the order of four modulo n is h, so its minimal polynomial over $\mathbb{F}_4$ is $X^h + \omega^2$. Thus a collision with $W < h$ would make the two polynomials identical. Their constant coefficients first force $s_A \equiv s_B \pmod 3$. Cancel that common scalar and let j_0 be the least index in $A \cup B$. If, say, $j_0 \in A$, then the coefficients of X^{j_0} on the two sides are respectively ω^{j_0} and 1. They differ because $3 \nmid j_0$; the other case is symmetric. More sharply, if $s_A \not\equiv s_B \pmod 3$, the difference has span W, while if $s_A \equiv s_B \pmod 3$, its constant and leading terms cancel but its coefficients at degrees j_0 and $W - j_0$ are nonzero, so its span is $W - 2j_0$. A nonzero multiple of $X^h + \omega^2$ has span at least h. This proves both pairwise criteria in the statement.

When $s_m < h$, necessarily $m < n$, so $\mathcal{A}_m \subseteq \mathcal{U}_h$. For $E, F \subseteq \mathcal{A}_m$, their symmetric difference has weighted sum at most s_m. Hence all 2^{t_m} displayed powers are distinct.

Every subset E with $s_E \le R_h$ belongs to a family whose pairwise weighted symmetric differences have sum at most $2R_h < h$. The number of such subsets is exactly $\mathcal{P}_h$, proving the first finite bound. For its asymptotic, put

$$G(t) = \prod_{j \ge 1} (1 + e^{-tj}), \qquad F(e^{-t}) = \prod_{3 \nmid j} (1 + e^{-tj}) = \frac{G(t)}{G(3t)}.$$

The standard q-Pochhammer asymptotic gives

$$G(t) \sim 2^{-1/2} \exp\left(\frac{\pi^2}{12t}\right), \qquad \frac{F(e^{-t})}{1 - e^{-t}} \sim t^{-1} \exp\left(\frac{\pi^2}{18t}\right).$$

The coefficients of $F(X)/(1 - X)$ are the nonnegative, nondecreasing cumulative counts in $\mathcal{P}_h$. Ingham's exponential Tauberian theorem [8] therefore gives

$$\log \mathcal{P}_h = 2\sqrt{\frac{\pi^2 R_h}{18}} + o(\sqrt{h}) = \frac{\pi}{3}\sqrt{h} + o(\sqrt{h}).$$

Finally, $t_m = 2m/3 + O(1)$ and $s_m = m^2/3 + O(m)$; taking the largest m with $s_m < h$ gives the elementary initial-segment estimate. The final assertion follows because a missing full ℓ-part would force $\operatorname{ord}(\eta_k) \mid N/\ell$. □

There is a second, sharp consequence of the same selected formula. It classifies the first possible one-sided relation rather than merely packing many pairwise-distinct powers.

Proposition 6.7 (sharp selected first-relation threshold)**.** *Retain the notation above. For every nonempty $E \subseteq \mathcal{U}_h$, put $S = s_E$ and decompose*

$$B_E(X) = \prod_{j \in E} (1 + X^j) = B_0(X^3) + X B_1(X^3) + X^2 B_2(X^3), \qquad B_a \in \mathbb{F}_2[Y].$$

If $a_0 \equiv -S \pmod 3$, then the exact all-weight criterion is

$$\eta_k^{Q_E} = 1 \quad \Longleftrightarrow \quad Y^{2H} + Y^H + 1 \mid B_a(Y) \quad \text{for both } a \ne a_0, \qquad H = h/3.$$

In particular,

$$\eta_k^{Q_E} = 1 \quad \Longrightarrow \quad s_E \ge 2h + 1.$$

The threshold is sharp: with $H = h/3$,

$$E_0 = \{1, h - 1, h + 1\}, \qquad Q_{E_0} = 1 + 2^H + 2^{2H} = |U_k|, \qquad s_{E_0} = 2h + 1.$$

Consequently, if a prime $\ell \mid N = |U_k|$ is absent from the selected order at its full valuation, write

$$\frac{N}{\ell} = \sum_{0 \le z < 2H} \epsilon_z 2^z, \qquad j_z \equiv 2^z \pmod{3h}, \quad 1 \le j_z < 3h.$$

Then necessarily

$$\sum_{\epsilon_z = 1} j_z \ge 2h + 1.$$

For this quotient word put $S = \sum_{\epsilon_z=1} j_z = 3T + 1$. Its two surviving sections are padded reciprocals:

$$B_1(Y) = Y^T B_0(Y^{-1}).$$

Thus the two divisibility conditions are equivalent, and the exact failure criterion is the single condition

$$\eta_k^{N/\ell} = 1 \quad \Longleftrightarrow \quad Y^{2H} + Y^H + 1 \mid B_0(Y).$$

Equivalently, define $\mathcal{T}_m \in \mathbb{F}_2[Y]$ by

$$\mathcal{T}_0 = \mathcal{T}_1 = \mathcal{T}_2 = 1, \qquad \mathcal{T}_m = \mathcal{T}_{m-1} + \mathcal{T}_{m-2} + (1+Y)\mathcal{T}_{m-3} \quad (m \ge 3).$$

Then, for $d = N/\ell$,

$$\eta_k^d = 1 \quad \Longleftrightarrow \quad Y^{2H} + Y^H + 1 \mid \mathcal{T}_d(Y).$$

Proof. Put

$$P_E(X) = \omega^S B_E(\omega X) + B_E(X).$$

The selected formula shows that $\eta_k^{Q_E} = 1$ exactly when $M(X) = X^h + \omega^2$, the minimal polynomial of w over $\mathbb{F}_4$, divides P_E. If $[X^r]B_E = b_r \in \mathbb{F}_2$, then

$$[X^r]P_E = b_r(1 + \omega^{S+r}).$$

Thus the scalar multiplier vanishes precisely on the section $r \equiv -S \pmod 3$. Write the three sections as in the statement. Since $h = 3H$, division by $M(X) = (X^3)^H + \omega^2$ respects these three sections. Divisibility by M therefore forces each of the two surviving $B_a(Y)$ to be divisible by $Y^H + \omega^2$. Because its coefficients lie in $\mathbb{F}_2$, conjugation also forces divisibility by $Y^H + \omega$, and hence by

$$(Y^H + \omega^2)(Y^H + \omega) = Y^{2H} + Y^H + 1.$$

The converse is immediate because the displayed polynomial contains $Y^H + \omega^2$ as a factor. This proves the exact all-weight criterion.

If $S < 2h$, every surviving section has degree below $2H$. If $S = 2h$, the top term lies in the vanished residue-zero section, so the same strict bound holds for the two surviving sections. Hence both are zero. The constant term of B_E then forces the sole supported section to be residue zero, but the least $j \in E$ contributes coefficient one at X^j, in a nonzero residue class. This contradiction proves $S \ge 2h + 1$.

The congruence $2^H \equiv h - 1 \pmod{3h}$ follows by induction from $2 \equiv 2 \pmod 9$ after cubing; squaring gives $2^{2H} \equiv h + 1 \pmod{3h}$. Hence E_0 has the asserted exponent, and $\eta_k^N = 1$ proves sharpness. Finally, N is odd and prime to three, so $\ell \ge 5$ and $N/\ell < 2^{2H}$. Its binary indices lie below $2H$ and have distinct residues modulo $3h$. A missing full ℓ-part gives $\eta_k^{N/\ell} = 1$, so the first assertion gives the displayed residue-weight bound.

For the quotient word, every current prime satisfies $\mathrm{ord}_\ell(2^H) = 3$, hence $\ell \equiv 1 \pmod 3$, while $N \equiv 1 \pmod 3$; therefore $d \equiv 1 \pmod 3$ and $S \equiv d \equiv 1 \pmod 3$. The product B_E is self-reciprocal of degree $S = 3T + 1$. If $[X^m]B_E = b_m$, then $b_m = b_{S-m}$, so comparison of the residue-zero and residue-one coefficients gives

$$B_1(Y) = Y^T B_0(Y^{-1}).$$

The polynomial $G(Y) = Y^{2H} + Y^H + 1$ is self-reciprocal. Since $\deg B_0 \le T$, reversing a factorization by G shows $G \mid B_0$ if and only if $G \mid B_1$; the zero and subdegree cases are included. The generic padded-mirror divisibility step is kernel-checked as `selfMirror_dvd_paddedMirror_iff`. The vanished section is $a_0 \equiv -S \equiv 2 \pmod 3$, proving the single-section criterion.

For the recurrence formulation, let y be a root of G, choose w with $w^3 = y$, and let $\omega^2+\omega+1 = 0$. The three elements $1 + w, 1 + \omega w, 1 + \omega^2 w$ are the roots of

$$Z^3 + Z^2 + Z + (1 + y),$$

so their power sums obey the displayed recurrence and have initial values $1, 1, 1$. Lucas's theorem gives $(1 + X)^d = \prod_{\epsilon_z=1}(1 + X^{2^z})$, and this product is congruent to $B_E(X)$ modulo $X^{3h} - 1$. Summing its values at the three cube roots of y projects onto the residue-zero section. Therefore $\mathcal{T}_d(y) = B_0(y)$. Since y has order $h = 3^k$ and $\mathrm{ord}_h(2) = 2H$, its minimal polynomial is G; the last divisibility criterion follows. □

The recurrence also admits a full symmetric composition law. This explains why the forced norm-triple relation at exponent N cannot simply be descended through $N = d\ell$.

Proposition 6.8 (cubic power-state composition saturation)**.** *Let K be a characteristic-two field containing $\omega^2 + \omega + 1 = 0$. Let L/K be a cyclic extension of degree three with generator σ, and let $w \in L^\times$ satisfy $\sigma(w) = \omega w$. Put $y = w^3 \in K$ and*

$$a = 1 + w, \qquad c = 1 + y, \qquad b = \frac{c}{a} = 1 + w + w^2,$$

and

$$A_m = \mathrm{Tr}_{L/K}(a^m), \qquad B_m = \mathrm{Tr}_{L/K}(b^m), \qquad C_m = c^m.$$

For universal variables define the sequence

$$D_0 = 1, \quad D_1 = A, \quad D_2 = A^2, \qquad D_n = AD_{n-1} + BD_{n-2} + CD_{n-3} \quad (n \ge 3).$$

Then, for all $m, n \ge 0$,

$$\begin{aligned} A_{mn} &= D_n(A_m, B_m, C_m), \\ B_{mn} &= D_n(B_m, A_m C_m, C_m^2), \\ C_{mn} &= C_m^n. \end{aligned}$$

If $m \equiv 1 \pmod 3$ and, for some positive integer H, the minimal polynomial of y over $\mathbb{F}_2$ is the self-reciprocal polynomial $G(Y) = Y^{2H} + Y^H + 1$, then

$$A_m = 0 \quad \Longleftrightarrow \quad B_m = 0.$$

Consequently, for every $n \equiv 1 \pmod 3$,

$$(A_m, B_m) = (0, 0) \quad \Longrightarrow \quad (A_{mn}, B_{mn}) = (0, 0).$$

The converse cannot hold uniformly: already at $H = 1$ one has $G \mid \mathcal{T}_7$ but $G \nmid \mathcal{T}_1$.

Proof. The eigenvalue relation makes $w \notin K$, so $K(w) = L$. It also gives $a \neq 0$; and $c = 0$ would make w a cube root of unity already contained in K, so $c \neq 0$. Thus all displayed quotients are defined. The three conjugates of a^m have elementary invariants (A_m, B_m, C_m). Newton's identities in characteristic two give the displayed recurrence for their power sums. The pairwise products have elementary invariants $(B_m, A_m C_m, C_m^2)$, which proves the second composition formula; the third is immediate.

Write $m = 3t + 1$ and decompose

$$(1 + X)^m = P_0(X^3) + XP_1(X^3) + X^2 P_2(X^3).$$

Self-reciprocity gives $P_1(Y) = Y^t P_0(Y^{-1})$. Since G is self-reciprocal, $A_m = P_0(y) = 0$ also gives $P_1(y) = 0$; hence $a^m \in w^2 K$. Therefore $b^m = c^m/a^m \in wK$ and $B_m = 0$. Conversely, decompose

$$(1 + X + X^2)^m = Q_0(X^3) + XQ_1(X^3) + X^2 Q_2(X^3).$$

This polynomial is self-reciprocal of degree $6t + 2$, so $Q_2(Y) = Y^{2t} Q_0(Y^{-1})$. The equality $B_m = Q_0(y) = 0$ therefore puts b^m in wK, and $a^m = c^m/b^m$ lies in $w^2 K$, giving $A_m = 0$.

When $A = B = 0$, the recurrence $D_n = CD_{n-3}$ has zero values in residue classes one and two modulo three. This proves the saturation statement. Finally, for $H = 1$ the recurrence gives $\mathcal{T}_1 = 1$ and $\mathcal{T}_7 = Y^2 + Y + 1 = G$. □

The first infinitesimal neighborhood of the selected factor is also exact. It does not contradict failure; instead it shows that any failure is transverse and recovers the unique Kummer preimage already forced by the projective equation.

Proposition 6.9 (selected cubic first-jet transversality)**.** *Use the notation of Proposition 6.8. Specialize to $k \geq 1$, $h = 3^k$, and $H = 3^{k-1}$, let y be the selected primitive h-th root whose minimal polynomial over $\mathbb{F}_2$ is $G(Y) = Y^{2H} + Y^H + 1$, and put*

$$q = 2^H, \qquad N = q^2 + q + 1, \qquad e = \frac{q+1}{3},$$

and let $\ell \mid N$ be prime, with $d = N/\ell$ and $s = (\ell - 1)/3$. Choose $w^3 = y$, and put $a = 1 + w$ and $c = 1 + y$. Then

$$\mathcal{T}_N(Y) = 1 + Y^e + Y^{qe}, \qquad \mathcal{T}_N'(y) = y^{e-1} \neq 0.$$

Suppose hypothetically that $G \mid \mathcal{T}_d$, and write $\mathcal{T}_d = GQ_d$. There is a unique λ with $a^d = w^2 \lambda$, necessarily with $\lambda \in K^\times$, and it is forced to be

$$\lambda = \frac{y^{e-1}}{c^{ds-1}}.$$

Moreover,

$$\mathcal{T}_d'(y) = \frac{y^{e-1}}{c^{ds}} \neq 0, \qquad Q_d(y) = c^{-ds}.$$

In particular $G^2 \nmid \mathcal{T}_d$. Equivalently, in the quotient by G^2 one has the exact first-order lift

$$\mathcal{T}_d(Y) \equiv G(Y)(1 + Y)^{-ds} \pmod{G(Y)^2},$$

where the negative power denotes the inverse of the unit $1 + Y$ in that quotient.

Proof. Lucas projection of

$$(1+X)^N = (1+X)(1+X^q)(1+X^{q^2})$$

onto exponents divisible by three gives the displayed formula for $\mathcal{T}_N$. The integer e is odd and qe is even, so its derivative at y is y^{e-1}.

Under the hypothetical zero state, the reciprocal-section argument in Proposition 6.8 puts a^d in w^2K, giving the unique λ. In the generic cubic algebra $w^3 = Y$, formal differentiation gives $dw/dY = w^{-2}$. Hence

$$\mathcal{T}_d'(y) = \mathrm{Tr}_{L/K}(a^{d-1}w^{-2}) = \lambda\,\mathrm{Tr}_{L/K}(a^{-1}) = \frac{\lambda}{c}.$$

Since $\ell = 3s+1$ and ℓ is odd, s is even. The universal cubic power sum has linearization

$$D_\ell(A,B,C) = AC^s + O((A,B)^2).$$

Differentiate the composition identity $\mathcal{T}_N = D_\ell(A_d, B_d, c^d)$ at the zero state. This gives

$$y^{e-1} = c^{ds}\mathcal{T}_d'(y),$$

which proves both formulas for λ and $\mathcal{T}_d'(y)$. Since $G'(y) = y^{H-1}$ and $e \equiv H \pmod h$, differentiating $\mathcal{T}_d = GQ_d$ at y gives $Q_d(y) = c^{-ds}$. The nonvanishing and the congruence modulo G^2 follow. □

For $k \geq 2$, the next Hasse jet also saturates. If $\epsilon_m = \binom{m}{2} \bmod 2$, direct differentiation under the same hypothetical failure gives

$$\mathcal{T}_d^{[2]}(y) = \frac{\lambda}{yc} + \epsilon_d\frac{\lambda}{c^2}, \qquad B_d'(y) = \frac{y\lambda^2}{c},$$

while

$$D_\ell(A,B,C) = AC^s + \epsilon_\ell B^2C^{s-1} + O((A,B)^3).$$

Because $N \equiv 1 \pmod 4$ and $d\ell = N$, one has $d \equiv \ell \pmod 4$ and hence $\epsilon_d = \epsilon_\ell$. The two quadratic terms cancel using $\lambda^3 = c^d/y^2$, leaving exactly $\mathcal{T}_N^{[2]}(y) = y^{e-2}$. Thus neither the ordinary derivative nor the second Hasse derivative supplies a character contradiction.

The two-base-q-digit degree route is arithmetically vacuous as well. Write $d = Aq + B$ with $0 \leq B < q$. Every current quotient satisfies

$$B > A, \qquad B - A \equiv 1 \pmod h, \qquad d > 1 \Longrightarrow B - A \geq h+1 \text{ and } B \geq h+2.$$

Indeed every prime factor of d is 1 modulo $2h$, while $q \equiv h-1 \pmod{2h}$; hence $d \equiv 1 \pmod{2h}$ and $B - A \equiv 1 \pmod h$. Also $d \mid N$ gives $d \mid A + q(A-B)$. Here $d < q^2$ gives $A < q$, while oddness of d gives odd $B \geq 1$. If $A \geq B$, then $0 < A + q(A-B) < Aq + B = d$, a contradiction. Thus $B > A$. If $d > 1$ and $B - A = 1$, then $A \geq 1$, and the same divisibility gives $d \mid q - A$ with $0 < q - A < d$, again a contradiction. Meanwhile the failure equation gives the nonzero polynomial

$$\mathcal{Q}(X) = (X+1)^B + \omega^{B+1}(X+\omega)^A(X+\omega^2)^{B-A}, \qquad \mathcal{Q}(w) = 0.$$

Its exponent span is B unless $A \equiv 1 \pmod 3$, when it is $B-2$. Divisibility by the degree-h selected factor therefore requires only $B \geq h$ or $B \geq h+2$, respectively. The digit span excludes the already-settled $d = 1$ prime-complement case directly; for every nontrivial current quotient $d > 1$, the quotient arithmetic already gives the stronger bound in both cases.

Thus every current $\ell \equiv 1 \pmod 3$ carries a hypothetical deficient d-state to the forced $N = d\ell$ state. In the finite-field specialization used below, $K = \mathbb{F}_{q^2}$ and $L = \mathbb{F}_{q^6}$. The relevant cyclic projective quotient has order

$$|L^\times/K^\times| = q^4 + q^2 + 1 = (q^2 + q + 1)(q^2 - q + 1),$$

and the two displayed factors are coprime. Hence for $\ell \mid q^2 + q + 1$ its ℓ-power map has kernel exactly ℓ. The forced projective equation

$$X^\ell = [w^{2\ell}]$$

therefore has exactly ℓ solutions, one of which is the deficient target $[w^2]$. This exact fibre belongs to the projective power map, not to the symmetric pair (A_d, B_d) itself. It proves that the symmetric recurrence and projective raising equation alone do not select the literal preimage; any resolvent or determinant refinement must supply an additional selector-sensitive ancestry input.

The all-weight criterion has an equivalent projective-torsion form which pinpoints why Hilbert–90 does not close it. This retains the marked root, rather than replacing it by a generic norm-one element.

Proposition 6.10 (marked projective obstruction and cohomological blindness)**.** *Put $H = h/3$, $q = 2^H$, $N = q^2 + q + 1$, and let $\ell \mid N$ be prime, with $d = N/\ell$. In*

$$L = \mathbb{F}_{2^{2h}}, \qquad K = \mathbb{F}_{q^2},$$

write $w = \zeta_k^2$ and

$$Z = \frac{(1+w)^d}{w^2}.$$

Then

$$\eta_k^d = 1 \quad \Longleftrightarrow \quad Z \in K.$$

Moreover, with $\alpha = w^3$,

$$Z^\ell = \omega^2(1+\alpha)\alpha^{-(2\ell+1)/3} \in K.$$

Thus $[Z] \in L^\times/K^\times$ is an explicitly marked ℓ-torsion class, and arm C asks that this class be nontrivial for every current ℓ.

Let $\sigma(x) = x^{q^2}$ generate $\mathrm{Gal}(L/K)$*. On μ_ℓ one has*

$$1 + \sigma + \sigma^2 = 0, \qquad \sigma - 1 \text{ is bijective}, \qquad H^1(\langle\sigma\rangle, \mu_\ell) = 0.$$

Consequently the norm and Hilbert–90 conditions on the displayed torsion class are automatic; they do not distinguish $[Z] = 1$ from a nontrivial marked class.

Proof. The congruences

$$2^H \equiv h - 1, \qquad 2^{2H} \equiv h + 1 \pmod{3h}$$

give

$$w^q = \omega^2 w^{-1}, \qquad w^{q^2} = \omega^2 w.$$

The selected formula in the proof of Proposition 6.6 gives

$$\frac{1 + \omega^2 w}{1 + w} = \omega \eta_k^{q+1}.$$

Since every prime divisor ℓ of N satisfies $\ell \equiv 1 \pmod 3$, also $d \equiv 1 \pmod 3$. Therefore

$$Z^{q^2-1} = \omega^2 \left(\frac{1+\omega^2 w}{1+w} \right)^d = \eta_k^{d(q+1)}.$$

As $\gcd(q+1, N) = 1$, this is one exactly when $\eta_k^d = 1$, which proves the fixed-field equivalence.

The three conjugates under q-powering satisfy

$$(1+w)^N = (1+w)(1+\omega^2 w^{-1})(1+\omega^2 w) = \frac{\omega^2(1+w^3)}{w}.$$

Hence

$$Z^\ell = \frac{(1+w)^N}{w^{2\ell}} = \omega^2(1+\alpha)\alpha^{-(2\ell+1)/3};$$

the exponent is integral because $\ell \equiv 1 \pmod 3$. The left side is fixed by q^2-powering (equivalently, this also follows by direct use of $\alpha^{q^2} = \alpha$), so it lies in K.

Finally put $Q = q^2$. Modulo ℓ,

$$1 + Q + Q^2 = 1 + q^2 + q^4 \equiv 1 + q^2 + q = 0,$$

because $q^3 \equiv 1$. On the other hand $\gcd(Q-1, \ell) = 1$: any common divisor of $q^2 - 1$ and $q^2 + q + 1$ divides three, whereas $\ell \neq 3$. Thus the norm operator is zero on μ_ℓ and the coboundary operator $\sigma - 1$ is an automorphism. The standard cyclic formula

$$H^1 = \ker(1 + \sigma + \sigma^2)/(\sigma - 1)\mu_\ell$$

is therefore zero. This proves the claimed blindness, not the desired pointwise nontriviality of $[Z]$.

More explicitly, $c = \sigma(Z)/Z$ lies in μ_ℓ because $Z^\ell \in K$. There is a unique $\rho \in \mu_\ell$ with $c = \sigma(\rho)/\rho$, and then $Z/\rho \in K$, so $[Z] = [\rho]$ in $L^\times/K^\times$. Every possible marked projective class is therefore represented by a coboundary; the zero cohomology class does not say whether its representing root ρ equals one. □

Together, the partition and first-relation bounds strengthen the polynomial Singer sieve to an exponential-in-$\sqrt{h}$ exclusion and a sharp weight-$2h$ exclusion for the actual norm-one coordinate. They remain subexponential and do not classify relations above the norm triple, so they do not force primitivity.

There is an exact number-field factorization of the remaining index. Put $L = \mathbb{Q}(\zeta_{3^{k+1}})^+$ and $K = \mathbb{Q}(\zeta_{3^k})^+$, reduce global units E_F and circular units C_F modulo the inert prime over two, and write

$$R_F = (\mathcal{O}_F/2)^\times/\overline{E}_F, \qquad T_F = \overline{E}_F/\overline{C}_F.$$

Then

$$[U_k : \langle \eta_k \rangle] = |\ker(\mathrm{N} : R_L \to R_K)| \frac{[\overline{E}_L : \overline{C}_L]}{[\overline{E}_K : \overline{C}_K]}.$$

The first factor is a relative principal 2-ray kernel; the second divides the relative real class-number quotient. Even class number one would leave the principal-ray factor. This is not a formal possibility only: analogous prime-conductor periods can have inert two and class number one while their finite circular-unit reduction has nontrivial defect [7]. Arm C is exactly the assertion that both factors in the displayed index formula vanish at every level.

7 The exceptional arm: corrected norm and Capelli

Let $h = 3^k$, $q = 2^h$, $\zeta = \zeta_k$, and $\omega = \zeta^h \in \mathbb{F}_4$.

Proposition 7.1 (first possible translate)**.** *If $f(p) = 2 \cdot 3^k$, then $m_p \geq 4$.*

Proof. Such a prime divides $2^h + 1$ but neither 3^{k+1} nor $2^h - 1$. The half-angle decomposition in Proposition 6.1 shows that each of $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$ has order prime to p. All four are p-th powers in their defining fields, so the excess definition gives the claim. □

The translate by four lies in a compositum with $\mathbb{F}_{16}$. The nontrivial relative Frobenius over $\mathbb{F}_4$ exchanges the nimbers 4 and 5, which are the roots of $Y^2 + Y + \omega$.

Proposition 7.2 (corrected norm)**.** *In $\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}$,*

$$\mathrm{N}(\zeta + 4) = (\zeta + 4)(\zeta + 5) = \zeta^2 + \zeta + \omega = A_k.$$

Moreover,

$$\mathbb{F}_2(\zeta + 4) = \mathbb{F}_{2^{4h}}.$$

Proof. One has $4 + 5 = 1$ and $4 \cdot 5 = \omega$. Expanding the product gives the formula.

Put $v = \zeta + 4$ and suppose $v^{2^j} = v$. Then

$$c := \zeta^{2^j} + \zeta = 4^{2^j} + 4$$

belongs both to $\mathbb{F}_{2^{2h}}$ and to $\mathbb{F}_{16}$. Since h is odd, their intersection is $\mathbb{F}_4$, so $c \in \{0, 1, \omega, \omega^2\}$. If $c \neq 0$, the Frobenius conjugate $\zeta^{2^j} = \zeta + c$ would have order $3^{k+1} \operatorname{ord}(\gamma_k) > 3^{k+1}$ by Proposition 6.1, whereas every Frobenius conjugate of ζ has order exactly 3^{k+1}. Hence $c = 0$. It follows simultaneously that $2h \mid j$ and $4 \mid j$. Their least common multiple is $4h$, so the Frobenius orbit of v has size $4h$, proving the generation assertion. □

Define

$$R_k(Y) = \mathrm{Res}_X \left(X^{2h} + X^h + 1,\ Y + X^h + X^2 + X \right).$$

Theorem 7.3 (exact exceptional target)**.** *The polynomial R_k is the irreducible degree-$2h$ minimal polynomial of A_k. The prime divisors of $\Psi_k = \Phi_{2 \cdot 3^k}(2)/3$ are exactly the primes $\ell \neq 3$ with $\operatorname{ord}_\ell(2) = 2h$, with their full valuations. The following are equivalent:*

1. *every row with $f(\ell) = 2 \cdot 3^k$ has $m_\ell = 4$;*
2. *$\Psi_k \mid \operatorname{ord}(M_k)$;*
3. *A_k is not an ℓ-th power in $\mathbb{F}_{2^{2h}}$ for every prime $\ell \mid \Psi_k$;*
4. *$R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.*

Proof. The element ζ has minimal polynomial $X^{2h} + X^h + 1$. To justify the degree of A_k, suppose $A_k^{2^j} = A_k$, put $t \equiv 2^j \pmod{3h}$, and write $u = \zeta^t$. If j is even, then $\omega^t = \omega$ and

$$(u + \zeta)(u + \zeta + 1) = 0,$$

so u is either ζ or $\zeta + 1$. If j is odd, then $\omega^t = \omega^2$ and

$$(u + \zeta + \omega)(u + \zeta + \omega^2) = 0,$$

so u is either $\zeta+\omega$ or $\zeta+\omega^2$. The three translated possibilities are impossible: Proposition 6.1 gives each the same order as $\zeta+1$, namely $3^{k+1}\operatorname{ord}(\gamma_k) > 3^{k+1}$; here $\operatorname{ord}(\gamma_k) > 1$ because $\mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}$ and $h > 1$. On the other hand, u is a 3^{k+1}-st root of unity. Hence $u = \zeta$, so $2^j \equiv 1 \pmod{3h}$. Since the order of two modulo $3h = 3^{k+1}$ is $2h$, the Frobenius orbit of A_k has size $2h$. Thus A_k has degree $2h$ and the displayed resultant is its minimal polynomial.

The cyclotomic divisor criterion applied to $\Phi_{2\cdot 3^k}(2)$ gives the stated prime support after removing its single factor of three. Proposition 7.1 and the corrected norm, now taken from the actual defining field of $\zeta+4$, give

$$m_\ell = 4 \quad\Longleftrightarrow\quad A_k \notin (\mathbb{F}_{2^{2h}}^{\times})^\ell.$$

Raising to $q-1$ is invertible on the current ℓ-Kummer quotient, so this is equivalent to the full ℓ-primary part occurring in $\operatorname{ord}(M_k)$. Finally, μ_ℓ lies in the defining field; Kummer theory and Capelli's lemma identify nonpower with irreducibility of $R_k(X^\ell)$. □

The exceptional Euler test also has a lossless sparse form parallel to the three-block criterion in the cubic arm.

Proposition 7.4 (sparse exceptional block criterion)**.** *Let $n = 3h$, let $\ell \mid \Psi_k$ be prime, put $d = (q+1)/\ell$, and work in $\mathcal{R}_h = \mathbb{F}_2[X]/(X^n-1)$. With x the class of X, set*

$$\mathcal{A} = x^2 + x + x^h, \qquad \iota(x) = x^{-1}, \qquad C = \mathcal{A}^d.$$

Write the unique cyclic representative of $V = C + \iota(C)$ as

$$V = V_0 + X^h V_1 + X^{2h} V_2, \qquad \deg V_i < h.$$

Then

$$M_k^{(q+1)/\ell} = 1 \quad\Longleftrightarrow\quad V_0 = V_1 = V_2.$$

If $d = \sum_{i=0}^{h-1} \epsilon_i 2^i$, then, losslessly in $\mathcal{R}_h$,

$$C = \prod_{\epsilon_i = 1} \left(x^{2^{i+1}} + x^{2^i} + x^{h2^i} \right),$$

where exponents are read modulo n. Thus arm D is equivalent to failure of this three-block equality for every current prime ℓ.

Proof. The order of two modulo $3^{k+1} = n$ is $2h$, so $q = 2^h \equiv -1 \pmod{n}$. Hence q-powering on $\mathcal{R}_h$ is ι. In the selected field $\mathbb{F}_2[X]/(J_h)$, where $J_h = X^{2h} + X^h + 1$, the image of $\mathcal{A}$ is A_k, and

$$M_k^d = 1 \quad\Longleftrightarrow\quad (A_k^d)^q = A_k^d \quad\Longleftrightarrow\quad J_h \mid C + \iota(C).$$

The first equivalence is kernel-checked abstractly as `exceptional_euler_fixed_iff`. Modulo J_h, characteristic two gives $X^{2h} = X^h + 1$, and therefore

$$V \equiv (V_0 + V_2) + X^h (V_1 + V_2).$$

The right side has degree less than $2h = \deg J_h$, so it vanishes exactly when $V_0 = V_1 = V_2$. The product formula is the binary-power expansion of $\mathcal{A}^d$ in characteristic two. □

Corollary 7.5 (primitive-only palindromicity)**.** *In the notation of Proposition 7.4, one always has*

$$V_0 + V_1 + V_2 = 0.$$

Consequently the following conditions are equivalent:

$$M_k^d = 1, \qquad V_0 = V_1 = V_2, \qquad V_0 = V_1 = V_2 = 0, \qquad C = \iota(C) \text{ in } \mathcal{R}_h.$$

In particular, every proper cyclic fold through $\mathbb{F}_2[X]/(X^h - 1)$ *vanishes identically on* V*; no Fourier test on a proper* 3*-power quotient can detect arm* D*.*

Proof. By lifting the exponent of three,

$$v_3(2^h + 1) = k + 1,$$

so $n = 3^{k+1}$ divides $q + 1$. Since $\ell \neq 3$, it follows from $q + 1 = \ell d$ that $n \mid d$. Modulo $X^h - 1$ one has

$$\mathcal{A} = X^2 + X + 1, \qquad \iota(\mathcal{A}) = X^{-2}\mathcal{A}.$$

Hence

$$\iota(C) = X^{-2d}C = C \pmod{X^h - 1}.$$

The image of $V_0 + X^h V_1 + X^{2h} V_2$ in this quotient is $V_0 + V_1 + V_2$, proving the first assertion. If the three blocks are equal to W, their sum is $3W = W$ in characteristic two, so $W = 0$. The converse is immediate, and every quotient $X^m - 1$ with $m \mid h$ factors through $X^h - 1$. □

There is a second, complementary compression which shortens the binary product by one third before testing this palindromicity.

Proposition 7.6 (one-third Frobenius compression)**.** *Put*

$$H = 3^{k-1}, \qquad h = 3H, \qquad Q = 2^H, \qquad S = Q^2 - Q + 1 = 3\Psi_k,$$

and, for a prime $\ell \mid \Psi_k$*, put* $e = S/\ell$*. Then*

$$Q \equiv h - 1 \pmod{3h}, \qquad S \equiv 3 \pmod{3h}, \qquad e \equiv 3 \pmod{3h},$$

and

$$d = (Q + 1)e.$$

In $\mathcal{R}_h$*, define*

$$\widetilde{\mathcal{B}} = \mathcal{A}^{Q+1}.$$

It has the lossless seven-term representative

$$\widetilde{\mathcal{B}} = 1 + x^{-2} + x^h + x^{h+1} + x^{2h} + x^{2h+1} + x^{2h+2},$$

and $C = \widetilde{\mathcal{B}}^e$*. In the primitive component* $\mathbb{F}_2[X]/(J_h)$ *this reduces further to the trinomial*

$$\mathcal{B} = x^{-2} + x + x^{2h+2}.$$

Thus

$$M_k^d = 1 \quad \Longleftrightarrow \quad \mathcal{B}^e = \iota(\mathcal{B})^e \text{ in } \mathbb{F}_2[X]/(J_h),$$

while arm D asks for the opposite inequality. The exponent now has at most $2H+1 = 2h/3+1$ binary places, rather than $h+1$.

If $e = 3$—equivalently, if $\Psi_k = \ell$ is prime—then the opposite inequality always holds. Hence every prime-complement level satisfies arm D; in particular this gives proof-level, non-enumerative certificates for $k = 2$ and $k = 3$.

More generally, put $r = \Psi_k/\ell$ and

$$\varepsilon_k = \begin{cases} 1, & k \text{ even}, \\ \omega^2, & k \text{ odd}. \end{cases}$$

For the reciprocal ratio

$$P = \frac{\iota(\mathcal{B})}{\mathcal{B}} = M_k^{Q+1}$$

one has the exact identities

$$P^{\Psi_k} = \varepsilon_k, \qquad M_k^d = 1 \quad \Longleftrightarrow \quad P^r = \varepsilon_k.$$

Thus, after removing its explicitly known three-primary phase, the unresolved composite target has exponent $r = \Psi_k/\ell \equiv 1 \pmod{2h}$ rather than $e = 3\Psi_k/\ell$.

Proof. The congruence for Q follows by induction. The base case is $2 \equiv 3 - 1 \pmod 9$. If $Q+1 = h(1+3t)$, write $h = 3s$ and cube. Then

$$Q^3 + 1 = 3h\left(1 + 3\left[t - s(1+3t)^2 + s^2(1+3t)^3\right]\right),$$

which is the required congruence at the next level. Substitution gives

$$S \equiv (h-1)^2 - (h-1) + 1 \equiv 3 \pmod{3h}.$$

Since $S = 3\Psi_k$ and $\ell \mid \Psi_k$, the integer e is divisible by three. Moreover $\operatorname{ord}_\ell(2) = 2h$ implies $h \mid \ell - 1$. Therefore $3h \mid (\ell-1)e$, and from $\ell e = S$ one gets $e \equiv S \equiv 3 \pmod{3h}$.

The factorization $Q^3 + 1 = (Q+1)S$ gives $d = (Q+1)e$. Frobenius and the first congruence give

$$\mathcal{A}^Q = x^{2h-2} + x^{h-1} + x^{2h}.$$

Multiplication by $\mathcal{A} = x^2 + x + x^h$ cancels the two copies of x^{2h-1} and gives the displayed seven terms. Modulo J_h, the three terms $1 + x^h + x^{2h}$ vanish and $x^{h+1} + x^{2h+1} = x$, leaving $\mathcal{B}$. The fixed-field equivalence is then Proposition 7.4 and Corollary 7.5 applied to $C = \widetilde{\mathcal{B}^e}$.

We next determine the three-primary phase. Since $(Q+1)\Psi_k = (q+1)/3$,

$$P^{\Psi_k} = M_k^{(q+1)/3} = A_k^{(q^2-1)/3} = \mathrm{N}_{\mathbb{F}_{4^h}/\mathbb{F}_4}(A_k).$$

The minimal polynomial of ζ over $\mathbb{F}_4$ is $X^h + \omega$. If $y^2 + y + \omega = 0$, its other conjugate is $y+1$, so the resultant formula for the norm gives

$$\mathrm{N}(A_k) = (y^h + \omega)((y+1)^h + \omega).$$

Reduction by $y^2 + y + \omega = 0$ gives $y^{15} = 1$ and

$h \bmod 15$	3	9	12	6
y^h	$\omega + \omega^2 y$	$\omega + \omega y$	$1 + \omega^2 y$	ωy.

The residues of 3^k modulo 15 run through $3, 9, 12, 6$. Using $y(y+1) = \omega$, the displayed product is therefore ω^2 in the first and third cases and 1 in the second and fourth. This proves $P^{\Psi_k} = \varepsilon_k$.

Now $e = 3r$. If $P^e = 1$, then P^r is a cube root of unity. Moreover

$$(P^r)^\ell = P^{\Psi_k} = \varepsilon_k.$$

Every current prime satisfies $\ell \equiv 1 \pmod 3$, so raising a cube root of unity to the ℓ-th power fixes it. Hence $P^r = \varepsilon_k$. The converse follows from $\varepsilon_k^3 = 1$. Finally, since every prime factor of r is current and hence congruent to one modulo $2h$, one has $r \equiv 1 \pmod{2h}$.

It remains to prove the prime-complement assertion, where $r = 1$. Put $z = \zeta$ and multiply numerator and denominator of P by z^2. Then

$$P = \frac{N}{D}, \qquad N = z^4 + z + \omega, \qquad D = 1 + z^3 + \omega^2 z^4,$$

and direct factorization over $\mathbb{F}_4$ gives

$$N + D = \omega(z + \omega^2)(z^3 + \omega^2), \qquad N + \omega^2 D = \omega^2(z+1)(z^3 + \omega).$$

If k is even, the first identity shows $P \neq 1$; if k is odd, the second shows $P \neq \omega^2$. Indeed z has order $3h \geq 27$, so neither z nor z^3 can equal the displayed cube root of unity. Thus $P \neq \varepsilon_k$, as required. The only earlier level has $\Psi_1 = 1$ and no current prime. □

The normalized phase has a substantially simpler Kummer representative. It also has an exact cubic ancestry law, although that law is provably blind to the newly born primary coordinate.

Proposition 7.7 (exceptional Cayley coboundary and norm boundary)**.** *Assume $k \geq 2$, retain the notation of Proposition 7.6, and put*

$$V_k = \frac{z^2 + \omega^2 z + 1}{z^2 + \omega z + 1} = \frac{\gamma_k + \omega^2}{\gamma_k + \omega}.$$

Put also $U = \varepsilon_k^{-1} P$. For every prime $\ell \mid \Psi_k$, the elements M_k, P, U, and V_k have the same ℓ-power/nonpower status. Equivalently, arm D is the assertion that V_k retains the full current Ψ_k-primary order.

Under the relative cubic norm

$$\mathrm{N}_k : \mathbb{F}_4(z)^\times \longrightarrow \mathbb{F}_4(z^3)^\times$$

one has

$$\mathrm{N}_k(V_k) = V_{k-1}^{-1}.$$

Thus the elements

$$E_k = \begin{cases} V_k, & k \text{ even}, \\ V_k^{-1}, & k \text{ odd} \end{cases}$$

are norm coherent. Nevertheless, at the birth level of every current ℓ, every fibre of N_k contains both ℓ-th powers and non-ℓ-th powers. Hence this complete norm ancestry does not determine the selected ℓ-symbol.

Proof. Put $H = 3^{k-1}$ and $Q = 2^H$. The relative Frobenius $\sigma : x \mapsto x^{Q^2}$ fixes $\mathbb{F}_4(z^3)$ and satisfies $\sigma(z) = \omega z$, because $Q^2 \equiv h + 1 \pmod{3h}$. All divisions below are legitimate. The roots of either quadratic defining V_k lie in $\mathbb{F}_{16}^\times$ and therefore have order dividing 15, whereas each of z, ωz, and $\omega^2 z$ has order $3h \geq 27$; the ratio defining P is a ratio of the nonzero elements $\iota(\mathcal{B})$ and $\mathcal{B}$. Direct factorization gives

$$\frac{\sigma(V_k)}{V_k} = \omega^2 P, \qquad \text{or equivalently} \qquad P = \omega V_k^{Q^2-1}.$$

The scalar is essential in this literal equality. For a current ℓ, $\mathrm{ord}_\ell(Q^2) = 3$, so $\gcd(\ell, Q^2 - 1) = 1$. Also $\ell \equiv 1 \pmod 3$, and hence $\omega = \omega^\ell$ is itself an ℓ-th power. Raising by $Q^2 - 1$ acts invertibly on the ℓ-Kummer quotient, so it preserves power/nonpower status and the generated ℓ-primary subgroup. Finally $P = M_k^{Q+1}$, and Q has order six modulo ℓ, so $\gcd(\ell, Q+1) = 1$. Also $U = \varepsilon_k^{-1} P$ with $\varepsilon_k \in \{1, \omega^2\}$, and $\varepsilon_k = \varepsilon_k^\ell$ is an ℓ-th power. This proves all four power/nonpower equivalences. The equality with the Möbius transform of $\gamma_k = z + z^{-1}$ follows after dividing the quadratics by z.

For the norm recurrence, direct multiplication in $\mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$ gives

$$\prod_{i=0}^{2} \big((\omega^i z)^2 + \omega^2(\omega^i z) + 1\big) = z^6 + \omega z^3 + 1,$$

and

$$\prod_{i=0}^{2} \big((\omega^i z)^2 + \omega(\omega^i z) + 1\big) = z^6 + \omega^2 z^3 + 1.$$

Their quotient is $V_{k-1}(z^3)^{-1}$, proving both the norm formula and the alternating normalization. The polynomial cross-product, the two three-factor products, the Möbius identity, and the scalar-corrected Kummer step are kernel-checked as

`exceptional_cayley_coboundary_cross`,
`exceptionalCayleyNum_cubic_product`,
`exceptionalCayleyDen_cubic_product`,
`exceptionalCayley_eq_traceMobius`,
`exceptional_scaled_coboundary_isPthPower_iff`.

It remains to prove the claimed blindness. Put $B = 4^H$. Then $\mathrm{ord}_\ell(B) = 3$, so

$$\ell \nmid B - 1, \qquad \ell \mid B^2 + B + 1.$$

The kernel of $\mathbb{F}_{B^3}^\times \to \mathbb{F}_B^\times$ is cyclic of order $B^2 + B + 1$. If g generates the top group, the kernel is generated by g^{B-1}. Because $\ell \nmid B - 1$, this generator maps nontrivially, and hence surjectively, to the order-ℓ quotient

$$\mathbb{F}_{B^3}^\times / (\mathbb{F}_{B^3}^\times)^\ell.$$

Multiplying any point of a fixed norm fibre by the norm kernel therefore realizes every ℓ-Kummer class in that same fibre. Strict ancestors have trivial ℓ-Kummer quotient, while later cubic norm maps are isomorphisms on that quotient because $\ell \neq 3$. Thus either birth class extends to a fully norm-coherent tail. This is an exact alternative-parent countermodel to norm-only induction, not a counterexample on the literal Conway-selected ancestry. □

The parity-twisted exponent reduction does not by itself settle composite Ψ_k. After putting $U = \varepsilon_k^{-1} P$, it says exactly that $U^{\Psi_k} = 1$ and that arm D asks for $U^{\Psi_k/\ell} \neq 1$ for every current prime

ℓ. Thus it isolates the prime-to-three current coordinate but does not prove that every one of its prime factors occurs in $\operatorname{ord}(U)$. Here the normalization uses $\Psi_k \equiv 1 \pmod 3$ and $r \equiv 1 \pmod 3$, the latter already following from $r \equiv 1 \pmod{2h}$.

The ordinary-arm ratio packing has an exceptional analogue, provided the multiplier roots of unity and the degree of the marked root are balanced by an intermediate constant field.

Proposition 7.8 (intermediate-field exceptional packing sieve)**.** *Assume $k \geq 2$, put $h = 3^k$, and let the prime ℓ divide Ψ_k. For $0 \leq j < k$, set*

$$m_j = 3^j, \qquad R_j = 3^{k-j}, \qquad s_j = \left\lfloor \frac{R_j - 1}{8} \right\rfloor$$

and define

$$\mathcal{S}_k = \max_{0 \leq j < k} \sum_{t=0}^{\min(2m_j, s_j)} 2^t \binom{2m_j}{t} \binom{s_j}{t}.$$

If the current ℓ-coordinate is deficient, then

$$\frac{\Psi_k}{\ell} \geq \mathcal{S}_k.$$

Consequently arm D holds at every current factor satisfying

$$\frac{\Psi_k}{\ell} < \mathcal{S}_k.$$

Moreover, if $H_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$ is binary entropy, then

$$\log_2 \mathcal{S}_k \geq c_0 \sqrt{h} - O(\log h), \qquad c_0 = \frac{1}{4} + \frac{2}{3} H_2\left(\frac{3}{8}\right) + \frac{3}{8} H_2\left(\frac{2}{3}\right) > 1.2306.$$

Thus this sieve proves an exponential-in-$\sqrt{h}$ large-factor range at every exceptional level, while remaining subexponential in h.

Proof. Write $z = \zeta$, $d = \Psi_k/\ell$, and regard the normalized phase as the rational function

$$U(T) = \varepsilon_k^{-1} \frac{\mathcal{N}(T)}{\mathcal{D}(T)}, \qquad \mathcal{N}(T) = T^4 + T + \omega, \qquad \mathcal{D}(T) = 1 + T^3 + \omega^2 T^4.$$

The polynomial $T^h + \omega$ is the minimal polynomial of z over $\mathbb{F}_4$. The order of four modulo $3h$ is h, and the map

$$t \longmapsto z^{4^t - 1}, \qquad 0 \leq t < h,$$

is a bijection from the Frobenius orbit indices to μ_h: every value is in μ_h, and injectivity follows from the orbit size. Hence the roots of $T^h + \omega$ are exactly cz, $c \in \mu_h$. Since U is defined over $\mathbb{F}_4$, deficiency $U(z)^d = 1$ implies

$$U(cz)^d = 1 \qquad (c \in \mu_h).$$

Fix j, abbreviate $m = m_j$, $R = R_j$, $s = s_j$, and put $E = \mathbb{F}_{4^m}$. LTE gives

$$v_3(4^m - 1) = 1 + v_3(m) = j + 1,$$

so $\mu_{3m} \subset E$. Since $E \subset \mathbb{F}_{4^h}$, $[E(z) : E] = h/m = R$. Also z^R has order $3m$ and lies in E, so the minimal polynomial of z over E is the binomial $T^R + z^R$.

Over $\mathbb{F}_4$ introduce the six monic quadratics

$$f_1 = T^2 + \omega T + 1, \quad f_2 = T^2 + \omega T + \omega, \quad f_3 = T^2 + \omega^2 T + 1,$$
$$f_4 = T^2 + T + \omega, \quad f_5 = T^2 + T + \omega^2, \quad f_6 = T^2 + \omega^2 T + \omega^2.$$

They are distinct and irreducible by direct inspection in $\mathbb{F}_4$. Up to nonzero scalar factors,

$$\mathcal{N}(T) = f_1(T) f_2(T), \qquad \mathcal{D}(T) \doteq f_3(T) f_4(T),$$

and

$$\mathcal{N}(\omega T) \doteq f_4(T) f_5(T), \qquad \mathcal{D}(\omega T) = f_2(T) f_6(T),$$

where $\doteq$ denotes equality up to a nonzero scalar. Since m is odd, $E \cap \mathbb{F}_{16} = \mathbb{F}_4$, so all six quadratics remain irreducible over E.

Choose representatives $\mathcal{C}$ for μ_{3m}/μ_3. Scaled factors belonging to different representatives are disjoint. Indeed, a common root of $f_a(cT)$ and $f_b(c'T)$ would give $c'/c \in \mathbb{F}_{16}^{\times} \cap \mu_{3m} = \mu_3$, hence $c = c'$ by the choice of representatives, and then $a = b$ because the six factors are distinct. In particular, for each $c \in \mathcal{C}$ the two elements

$$u_{c,0} = U(cz), \qquad u_{c,1} = U(\omega c z)$$

have, up to scalars, the factor patterns

$$u_{c,0} \doteq \frac{f_1(cz) f_2(cz)}{f_3(cz) f_4(cz)}, \qquad u_{c,1} \doteq \frac{f_4(cz) f_5(cz)}{f_2(cz) f_6(cz)}.$$

Their denominators are nonzero: otherwise a conjugate of z would be a root over $\mathbb{F}_4$ of the degree-four polynomial $\mathcal{D}$, whereas z and all its conjugates have degree $h \geq 9$. All $2m$ displayed elements are d-th roots of unity.

Index them by a set I of size $2m$. For $v \in \mathbb{Z}^I$ with $\sum_i |v_i| \leq s$, put $V_v = \prod_i u_i^{v_i}$. Suppose $V_v = V_w$ and set $g = v - w$. Separating positive and negative exponents and cross-multiplying gives an equality at z between two polynomials in $E[T]$, each of degree at most

$$4 \sum_i |g_i| \leq 8s < R.$$

The degree-R minimal polynomial therefore forces equality of the two polynomials. Unique factorization now recovers every exponent: for a fixed c, the valuation at $f_1(cT)$ gives $g_{c,0} = 0$, and the valuation at $f_5(cT)$ gives $g_{c,1} = 0$. Cross-star disjointness handles all $c \in \mathcal{C}$. Thus the V_v are distinct.

The signed ℓ^1 ball in $\mathbb{Z}^{2m}$ has exactly

$$\sum_{t=0}^{\min(2m,s)} 2^t \binom{2m}{t} \binom{s}{t}$$

points. Since their images are distinct roots of $X^d - 1$, this number is at most d. Taking the maximum over j proves the exact bound.

For the asymptotic statement, first let k be even and take $j = k/2 - 1$. Then, with $x = \sqrt{h}$,

$$2m = \frac{2}{3}x, \qquad s = \frac{3}{8}x + O(1).$$

The summand with $t = \lfloor x/4 \rfloor$ and Stirling's formula give

$$\log_2 \mathcal{S}_k \geq \left[\frac{1}{4} + \frac{2}{3}H_2\left(\frac{3}{8}\right) + \frac{3}{8}H_2\left(\frac{2}{3}\right)\right] x - O(\log h).$$

If k is odd, take $j = (k-3)/2$. Then

$$2m = \frac{2}{3\sqrt{3}}x, \qquad s = \frac{3\sqrt{3}}{8}x + O(1),$$

and the same choice $t = \lfloor x/4 \rfloor$ gives coefficient

$$\frac{1}{4} + \frac{2}{3\sqrt{3}}H_2\left(\frac{3\sqrt{3}}{8}\right) + \frac{3\sqrt{3}}{8}H_2\left(\frac{2}{3\sqrt{3}}\right) > 1.2341 > c_0.$$

The $O(\log h)$ errors are the standard logarithmic errors in the two binomial-coefficient estimates. This proves the uniform claim for all sufficiently large k; enlarging the implicit constant absorbs the finitely many smaller levels. □

This packing theorem is a genuine composite-Ψ_k range, not a universal proof: its lower bound is exponential in $\sqrt{h}$, whereas Ψ_k is exponential in h. Logarithmic or Hasse derivatives do not add an intrinsic constraint in this finite etale quotient: $\mathbb{F}_2[X]/(J_h)$ has no nonzero $\mathbb{F}_2$-derivations, while the formal polynomial derivative does not descend because $J_h' = X^{h-1}$ is a unit modulo J_h. A derivative argument would therefore need genuinely new lift or multiplicity information. Proposition 7.6 and Proposition 7.8 therefore prove the prime-complement case and a large-factor range, not universal arm D.

Proposition 7.9 (effective exceptional bound)**.** *Let $k \geq 2$, $h = 3^k$, and let $\ell \mid \Psi_k$ be prime. If t is the least power of two satisfying $t \geq \lceil \log_2(h-1) \rceil$, then*

$$m_\ell \leq 4^t - 1 < 16h^4.$$

Proof. Put $K = \mathbb{F}_{4^h}$ and extend constants to $L = K\mathbb{F}_{4^t}$. For a character χ of exact order ℓ, the sum

$$S = \sum_{c \in \mathbb{F}_{4^t}} (\chi \circ \mathrm{N}_{L/K})(\zeta + c)$$

equals 4^t precisely when every translate is an ℓ-th power. The associated primitive even function-field character has L-polynomial $(1-T)\prod_{j=1}^{h-2}(1-\alpha_j T)$ with $|\alpha_j| = 2$. Constant-field extension therefore gives

$$|S| \leq 1 + (h-2)2^t < 4^t$$

when $2^t \geq h - 1$; see [16]. Some translate is a nonpower, and norm injectivity on the ℓ-Kummer quotient returns the statement to its defining field. The elements of $\mathbb{F}_{4^t}$ are the finite nimbers below 4^t. The displayed polynomial estimate follows from the minimal choice of t. □

This bound is level-dependent and does not prove absolute boundedness or the value four.

The exceptional coordinate also has an exact circular-ray factorization. In the real field generated by roots of orders 3^{k+1} and 5, let $\mathfrak{P}$ be the prime selected by $\zeta \mapsto \zeta_k$ and the conductor-five coordinate $\mapsto \omega$. Let E_L and C_L be the global and circular units and let a subscript ℓ denote the primary projection after reduction. Then

$$\overline{C}_{L,\ell} = \langle A_{k,\ell} \rangle = \langle M_{k,\ell} \rangle,$$

and

$$[k_{\mathfrak{P},\ell} : \langle M_{k,\ell} \rangle] = \left|\ker\big(\mathrm{Cl}_L(\mathfrak{P})_\ell \to \mathrm{Cl}_{L\,\ell}\big)\right| \, |(\overline{E}_L/\overline{C}_L)_\ell|.$$

The second factor is controlled by the real class number, but the first is a principal ray kernel. The marked conductor-five cyclotomic unit is globally not an ℓ-th power; the height theorem of Amoroso–Dvornicich gives this directly [1]. Global nonpower does not determine its Frobenius at the selected prime. Arm D is exactly the assertion that both factors in the displayed ray-class index vanish.

The current C and D coordinates are orthogonal. If $q_0 = 2^{h/3}$ and $q = q_0^3$, their norm-one tori have orders

$$q_0^2 + q_0 + 1 \quad \text{and} \quad q + 1,$$

which are coprime. The exceptional cubical operator kills the cubic subfield, while the local norm kills the exceptional torus. Hence the two principal-ray factors cannot force one another through the shared prime over two. Each requires its own marked unit-image calculation.

Even a perfect one-dimensional Kummer–Artin pairing does not show that a prescribed prime has nonzero image in the Artin quotient. The obstruction already occurs arithmetically.

Proposition 7.10 (perfect-pairing no-go)**.** *There exist a number field K containing μ_3, a global unit u which is not a cube in K, and exactly one prime $\mathfrak{q}$ of K above two; this prime splits completely in the nontrivial cyclic Kummer extension $K(\sqrt[3]{u})/K$.*

Proof. Take

$$K = \mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\zeta_3, \sqrt{3}), \qquad u = 2 + \sqrt{3}.$$

The Pell equation gives $\mathcal{O}^\times_{\mathbb{Q}(\sqrt{3})} = \{\pm u^n : n \in \mathbb{Z}\}$, so u is not a cube in the real quadratic subfield. If $X^3 - u$ acquired a root in its quadratic extension K, its cubic factorization over the real subfield would also have a linear factor. Hence u is not a cube in K, and adjoining a cube root gives a nontrivial cyclic cubic extension.

Locally at two, $\mathbb{Q}_2(\zeta_3)/\mathbb{Q}_2$ is unramified quadratic and adjoining i is ramified quadratic. The two local quadratics are linearly disjoint, so $K \otimes \mathbb{Q}_2$ is a degree-four field; hence two has a unique prime $\mathfrak{q}$ in K. In its residue field $\sqrt{3} \equiv 1$, so $u \equiv 1 \pmod{\mathfrak{q}}$. The polynomial $X^3 - u$ has the simple root 1 modulo $\mathfrak{q}$, because its derivative at one is the unit 3. Hensel's lemma makes u a cube in $K_\mathfrak{q}$; since $\mu_3 \subset K_\mathfrak{q}$, the polynomial splits completely there and $\mathfrak{q}$ splits completely in $K(\sqrt[3]{u})$. □

Thus nontriviality of the Kummer line, one-dimensionality of its dual, and even uniqueness of the prime above two are compatible with zero marked pairing. In the four arms, perfect duality reduces the target to nonvanishing of the Conway-selected prime class; it does not prove that nonvanishing.

8 Nim reciprocity saturation and marked holonomy

The residue-symbol reductions suggest using reciprocity to evaluate the remaining phase. There is a precise saturation boundary for that program. Let ℓ be an odd prime, let $s \in \mu_\ell$, and put $E = \mathrm{ord}_\ell(2)$. The two-axis Frobenius–cyclotomic family is $s_{j,c} = s^{2^j c}$ for $0 \le j < E$ and $c \in \mathbb{F}_\ell^\times$.

Proposition 8.1 (reciprocity saturation)**.** *The following statements hold.*

1. *Every multiplicative word in the $s_{j,c}$ is s^λ, where*

$$\lambda = \sum_{j,c} w_{j,c} 2^j c \pmod{\ell}.$$

It is blind when $\lambda = 0$ and, when $\lambda \neq 0$, equals one exactly when the original marked phase equals one.

2. *For multiplicative characters χ, ψ of a finite field and $a \neq 0$, the Jacobi convolution*

$$J_{\chi,\psi}(a) = \sum_x \chi(x)\psi(a-x)$$

satisfies

$$J_{\chi,\psi}(a) = \chi(a)\psi(a)J_{\chi,\psi}(1).$$

In particular, if both characters are powers of one order-ℓ character, the marked dependence is again one power of the same phase.

3. *In the affine group on $\mathbb{F}_\ell$, write (t, q), with $q \in \mathbb{F}_\ell^\times$, for the map $x \mapsto qx + t$. If $q \neq 1$, all (t, q) are conjugate. Consequently every class function is independent of t.*

Proof. The first statement follows by collecting exponents. For the second, substitute $x = ay$ and factor $\chi(a)\psi(a)$ from every summand. For the third, conjugation by the translation $x \mapsto x + a$ changes t to $t + (1 - q)a$; taking $a = t/(q - 1)$ makes this zero. □

Thus norm products, weighted Stickelberger words, and Gauss or Jacobi transforms either erase or re-encode one phase. A class function on the single fixed-multiplier affine coset is blinder still: it forgets the translation coordinate in part 3. None of these operations independently evaluates the selected phase.

Ordinary cross-arm multiplicative words face the same boundary. If a strict ancestor lies in a field $\mathbb{F}_{2^d}$ with $f(\ell) \nmid d$, exactness of the residue degree gives $\ell \nmid 2^d - 1$; the ℓ-power map is therefore an automorphism on that ancestor group and its ℓ-Kummer image is trivial. Thus every cross-arm word built from strict lower-level coordinates in such degrees separates its coprime-primary torsion. Nor does the next ℓ-component give an induction: its defining radicand is $\alpha_\ell = \kappa_{f(\ell)} + m_\ell$, so irreducibility proves only that the actual winning translate is a non-ℓ-th power. Using that fact to identify m_ℓ with the conjectural 0, 1, or 4 would already assume the target.

The intrinsic coordinate is a full Frobenius-orbit holonomy, not a one-step affine translation. Let $\beta \in \mathbb{F}_{2^E}^\times$ be the selected element, put $s = \beta^{(2^E-1)/\ell}$, choose $\alpha^\ell = \beta$ in an algebraic closure, fix a primitive $\zeta \in \mu_\ell$, and let $\varphi(z) = z^2$. Then

$$\frac{\varphi^E(\alpha)}{\alpha} = \alpha^{2^E-1} = \beta^{(2^E-1)/\ell} = s.$$

This ratio is unchanged when α is replaced by $\zeta^a\alpha$, because φ^E fixes μ_ℓ. If $s = \zeta^T$, the E-step action on the labeled roots is the pure translation $e_x \mapsto e_{x+T}$. Hence on the Fourier lines

$$v_r = \sum_x \zeta^{-rx} e_x, \qquad \varphi^E(v_r) = \zeta^{rT} v_r.$$

The diagonal eigenvalue at $r = 1$ is exactly the selected Kummer–Artin symbol.

Equivalently, choose a root frame over each conjugate $\beta_j = \beta^{2^j}$ and write the one-step maps as $e_{j,x} \mapsto e_{j+1,2x+t_j}$. Their cyclic holonomy is

$$T = \sum_{j=0}^{E-1} 2^{E-1-j} t_j \pmod{\ell}.$$

Changing frames by $\alpha_j \mapsto \zeta^{a_j}\alpha_j$ sends t_j to $t_j + 2a_j - a_{j+1}$; with $a_E = a_0$, these changes telescope because $2^E = 1$ modulo ℓ. Thus T is intrinsic, while the individual off-diagonal coefficients

$$\rho(t_j, 2)v_r = \zeta^{r2^{-1}t_j} v_{r2^{-1}}$$

are gauge-dependent. Every proper edge can be normalized away recursively, but a closed gauge cannot erase the cyclic holonomy: all but one edge may be made trivial, with the remaining edge carrying T.

This also explains the cocycle no-go. A homomorphic splitting on a commutative multiplier subgroup obeys $t(qr) = t(q) + qt(r)$. Fixing $q_0 \neq 1$ and comparing $t(qq_0)$ with $t(q_0q)$ gives

$$t(q) = (1-q)a, \qquad a = \frac{t(q_0)}{1-q_0}.$$

Every such cocycle is a coboundary and has trivial full-period transfer. Powers of one affine lift give this law on the subgroup generated by its multiplier. A general labeled section has a factor set, but changing root frames changes that factor set by a coboundary; only its cyclic holonomy is gauge-invariant.

Affine ancestry between two such torsor paths cannot create a missing phase. If the vertex maps are $x \mapsto \lambda x + b_j$, then the target translations have the form

$$u_j = \lambda t_j + b_{j+1} - 2b_j,$$

and the path holonomies satisfy

$$U_n = \lambda T_n + b_n - 2^n b_0.$$

On a closed E-cycle the endpoint term telescopes, so $U = \lambda T$. Consequently an affine recursion from a strict ancestor with trivial ℓ-holonomy cannot prove a nonzero current phase; a successful relation must contain a genuinely non-affine or independently nonzero closing source.

A successful nim-reciprocity proof must therefore retain and independently evaluate the marked holonomy T, equivalently s, along the literal Conway ancestry. Constructing the holonomy from the selected phase is an exact reformulation, not yet its evaluation.

9 Boundedness and the remaining theorem

Lenstra asked whether the finite excesses are absolutely bounded [9]. Conjecture 1.1 would give the bound four, but boundedness is strictly weaker than the maximal-order assertions above.

For $a \geq 0$, let $C_a = \mathbb{F}_{2^{2^a}}$. For an odd prime p, put $x_p = \kappa_{f(p)}$ and $L_{p,a} = \mathbb{F}_2(x_p)C_a$.

Proposition 9.1 (finite-line criterion)**.** *For every p and a,*

$$m_p < 2^{2^a} \quad \Longleftrightarrow \quad \textit{some } c \in C_a \textit{ has } x_p + c \notin L_{p,a}^{p}.$$

Consequently,

$$\sup_p m_p < \infty \quad \Longleftrightarrow \quad \textit{one fixed } a \textit{ works for every } p.$$

Proof. For $F_c = \mathbb{F}_2(x_p + c)$, one has $F_cC_a = L_{p,a}$ and $[L_{p,a} : F_c] \mid 2^a$, which is prime to p. Extension is therefore injective on p-Kummer quotients. The finite nimbers below 2^{2^a} are exactly C_a, proving the first equivalence from the excess definition. A fixed a gives a uniform bound; conversely choose a above any given uniform bound. □

No fixed finite affine line is known to work for every prime. The complete universal frontier remains the four selected order assertions in Theorem 3.1:

1. full primary order for the synchronized structural norm, including every Conway–Fermat equality $\delta_n = F_n$;
2. full primary projective order for the marked ordinary cyclotomic value;
3. primitivity of every selected Gaussian period γ_k, equivalently generation by every η_k;
4. $\Psi_k \mid \mathrm{ord}(M_k)$ at every exceptional level.

Each is now one-dimensional and ancestry-sensitive. Degree, trace, norm, conductor, factor parity, or a statement about generic ambient points does not decide it.

10 Authority and verification

The claims in this paper have six distinct sources. Table 1 is part of the theorem statement: in particular, a finite calculation or a Lean definition of a target is never used as evidence for a universal claim.

Status	Content
Cited input	Conway–Lenstra tower formulas, standard finite-field and Capelli theory, and the cited cyclotomic, class-field, function-field, and order results. These are not reproved in Lean.
Proved here	The four-arm equivalence; the exact selected coordinates and order criteria; the primitive-support, packet-degree, Wiedemann-sibling, supersingular, Singer and selected norm-one ratio-packing and partition and single-section first-relation bounds, corrected-norm, ordinary section, reciprocal-doubling, and unconditional intermediate-field Kummer-ratio selected-order bounds, exceptional palindromicity, one-third compression, intermediate-field packing, prime-complement exceptional case, ray-index, boundedness, and reciprocity-saturation results stated above.
Ordinary kernel-checked Lean	Algebra without native evaluation: the algebraic Kummer interface, full-primary power tests, selected cubic and exceptional identities, low-degree minimal-polynomial collision and normalized rational-packing injection and the cubic factor-of-three order bridge, cubical reductions, packet and Fibonacci identities, and marked-holonomy calculus. These proofs may still use Lean's standard logical dependencies `propext`, `Classical.choice`, and `Quot.sound`.
Native evaluation	Named finite arithmetic propositions proved with `native_decide`. Lean's axiom report records the generated `_native.native_decide.ax_1_1` for these declarations, so they are exact replayable finite calculations, not ordinary kernel reductions.
External evidence	Source-backed rows, Python certificates, and explicit countermodels to overstrong intermediate lemmas. These establish only their stated finite instances.
Open	The universal assertions Z, O, C, D, Conjecture 1.1, and absolute boundedness of m_p.

Table 1: Claim authority for the finite-excess paper.

The Lean 4 [5] development is exposed by the import-only entry point `Ogdoad.Papers.Excess` and uses Mathlib [18]. Its certificate namespaces share the single evaluator `Ogdoad.Certificate.BinaryPolynomial`; the moduli, field elements, and propositions remain local to the certificate they

identify. The identifier `DPrimeTarget`, like the analogous selected targets, merely defines an open proposition. It is neither an axiom nor a theorem. The formalization therefore verifies ingredients of Theorem 3.1, not an end-to-end proof of Conjecture 1.1.

The maintained finite record is as follows.

- The 126 values for odd primes $3 \le p \le 709$ are source-backed by OEIS A380496 [13]; they are data, not an induction.
- The rows $m_{359} = m_{719} = m_{727} = 1$ have separate exact local finite-field certificates.
- In the cubic arm, C_1, C_2, C_3 have exact order proofs and C_4, C_5 have complete finite certificates; C_6 retains a source-assisted factorization boundary.
- In the exceptional arm, D'_2, D'_3 follow analytically and D'_4, D'_5 have complete finite certificates; D'_6 retains a source-assisted primality boundary.
- Two-block tensor arguments prove the multicomponent zero arm at $h = 12, 15, 24, 36, 40$. They are theorem-level finite cases, not a universal generation theorem.

An explicit countermodel in a non-Conway field can refute a proposed proof from trace, norm, conductor, or factor shape. It is not a counterexample to the 0/1/4 rule unless it occurs on the literal recursively selected Conway ancestry.

11 Conclusion

The finite-excess problem is now four exact selected-order problems. The zero arm needs a synchronized primitive-support coordinate, the ordinary arm one Artin symbol on the Conway-selected inert ray, and the cubic and exceptional arms the faithful finite reductions of two marked units. Every reduction retains the full primary valuation; none replaces the selected coordinate by a generic element of the same field.

This is also the obstruction to the presently available proof routes. Trace, norm, conductor, class group, and reciprocity packages can reconstruct or transport the marked phase, but the saturation theorem shows that homogeneous transforms of that same phase do not evaluate it. A proof of the 0/1/4 rule must add an ancestry identity with independent arithmetic content. Until that identity closes all four arms, both the rule and absolute boundedness remain open.

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